



US012416381B2

(12) **United States Patent**
Keough et al.

(10) **Patent No.:** **US 12,416,381 B2**

(45) **Date of Patent:** **Sep. 16, 2025**

(54) **ADHESIVE TRANSFER HOSE HAVING A BARRIER LAYER AND METHOD OF USE**

(58) **Field of Classification Search**

CPC ... F16L 53/38; B32B 1/08; B32B 3/28; B32B 5/024; B32B 5/26

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(Continued)

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(*) Notice: Subject to any disclaimer, the term of this
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(21) Appl. No.: **18/632,366**

(22) Filed: **Apr. 11, 2024**

(65) **Prior Publication Data**

US 2024/0255089 A1 Aug. 1, 2024

Related U.S. Application Data

(63) Continuation of application No. 15/632,371, filed on
Jun. 25, 2017, now Pat. No. 12,031,658.
(Continued)

(51) **Int. Cl.**
F16L 53/38 (2018.01)
B32B 1/08 (2006.01)
(Continued)

(52) **U.S. Cl.**
CPC **F16L 53/38** (2018.01); **B32B 1/08**
(2013.01); **B32B 3/28** (2013.01); **B32B 5/024**
(2013.01);
(Continued)

Primary Examiner — Steven W Crabb
Assistant Examiner — Thomas J Ward

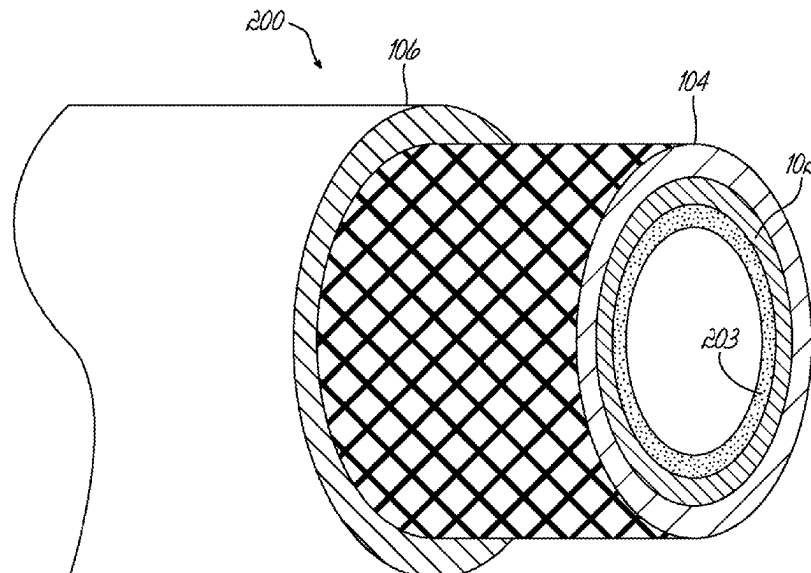
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(57)

ABSTRACT

A multi-layered hot melt adhesive transfer hose is provided. The transfer hose has a barrier layer that prevents or minimizes the ingress of oxygen and other gases into a conduit of the hose. The transfer hose has at least one structural layer overlaying an exterior surface of the barrier layer to help the hose withstand a high fluid pressure. The barrier layer prevents the hot melt adhesive being transported in the hose from discoloring and charring when the adhesive is heated from about 250° F. up to and including 450° F. for an extended period of time.

20 Claims, 18 Drawing Sheets



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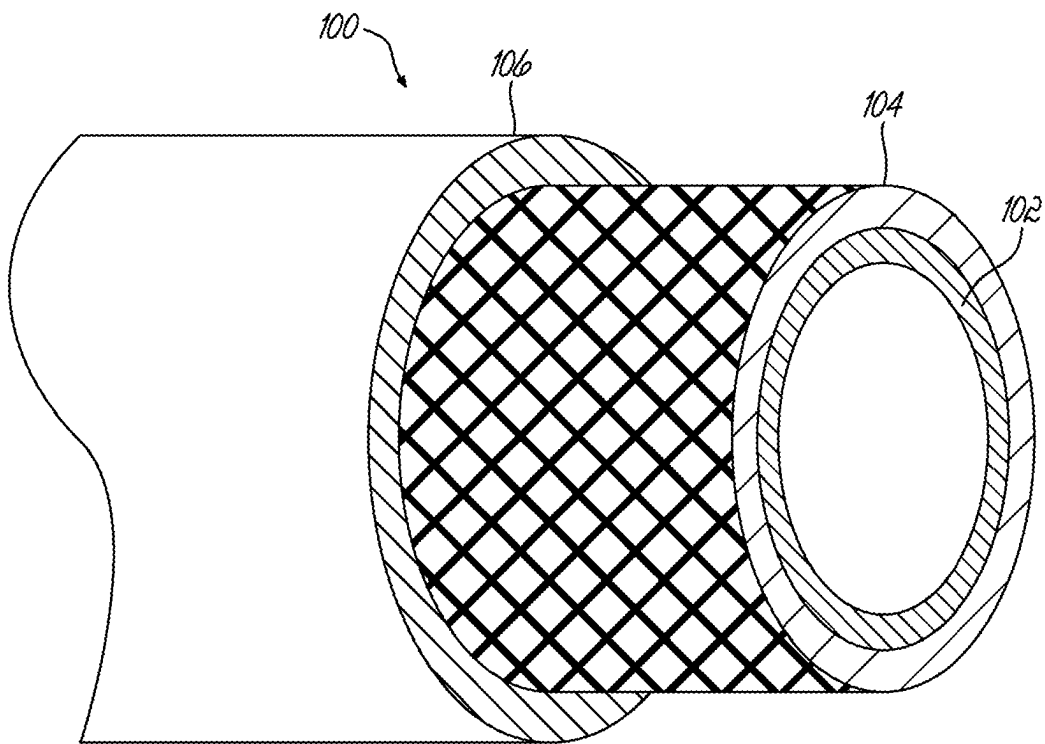


FIG. 1A
PRIOR ART

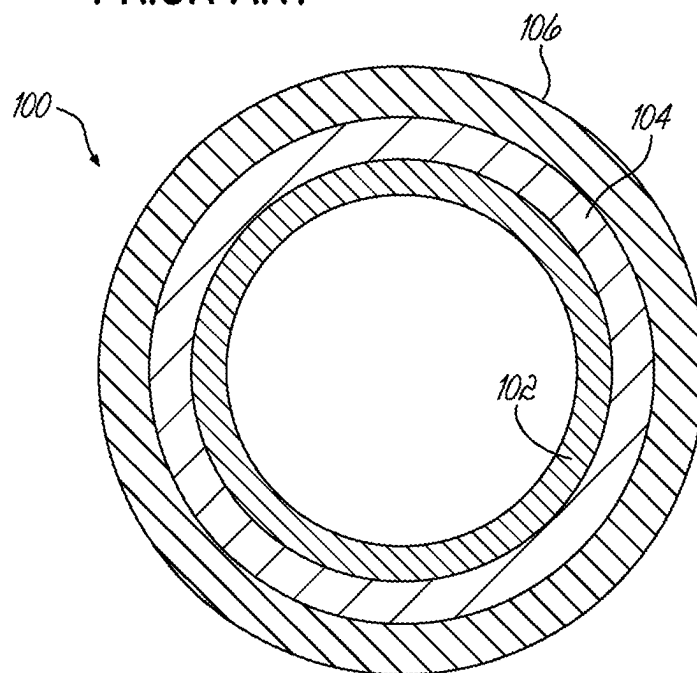


FIG. 1B
PRIOR ART

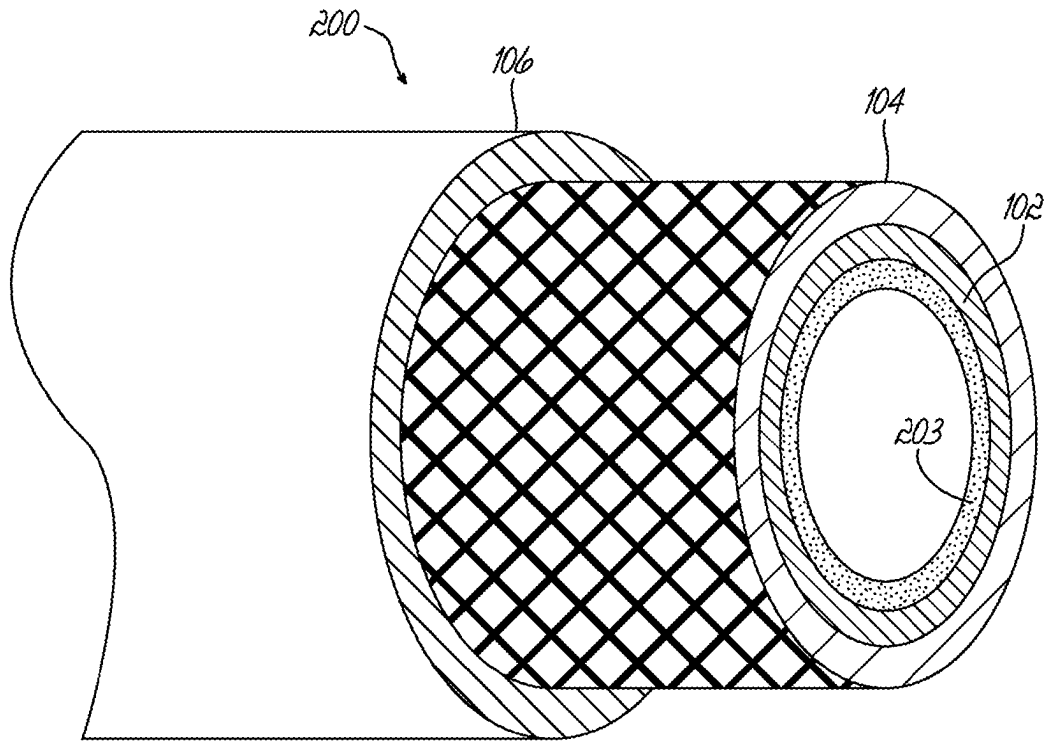


FIG. 2A

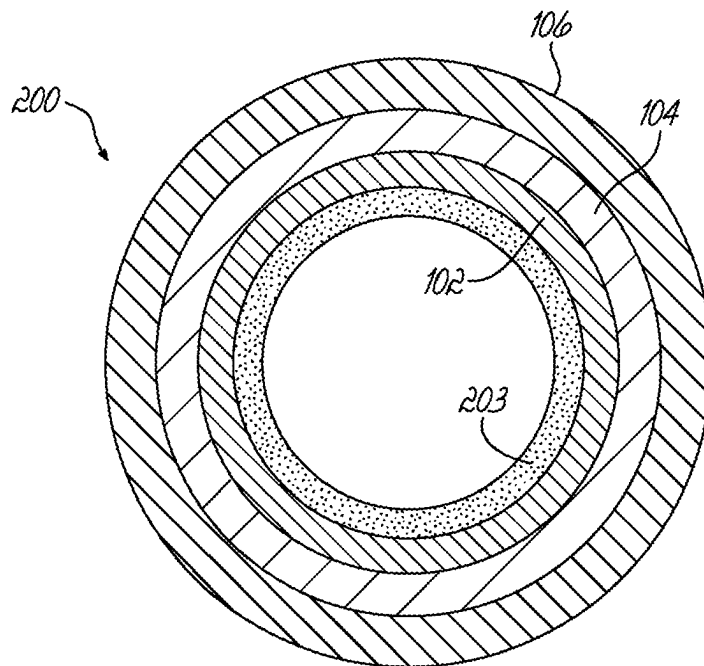


FIG. 2B

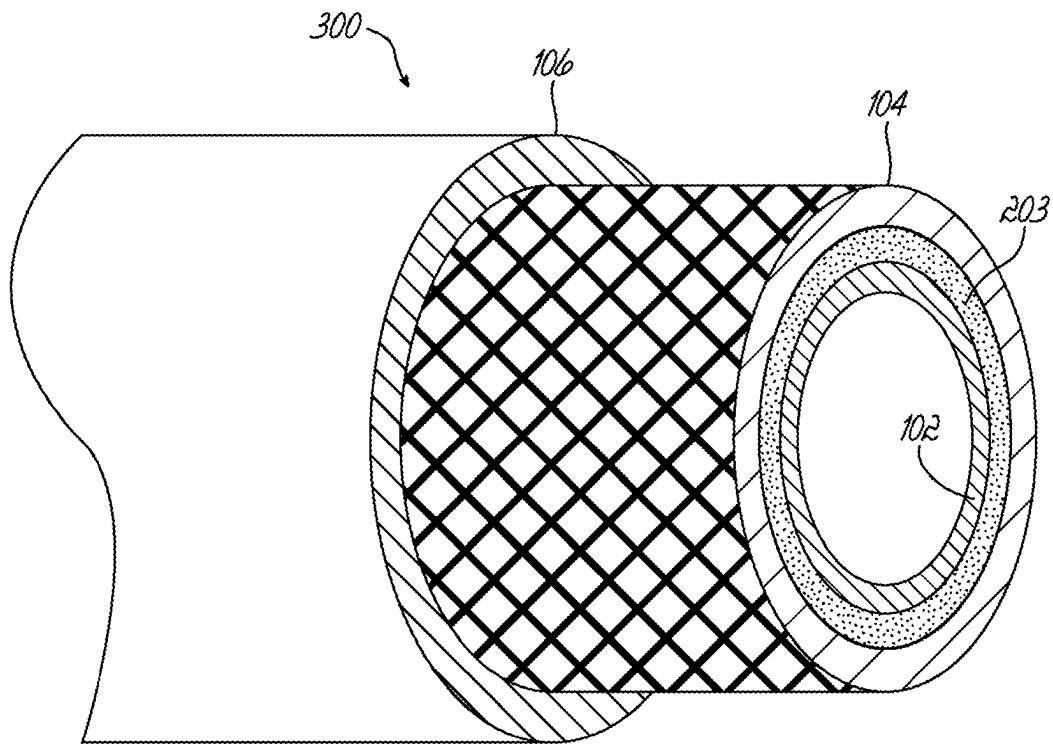


FIG. 3A

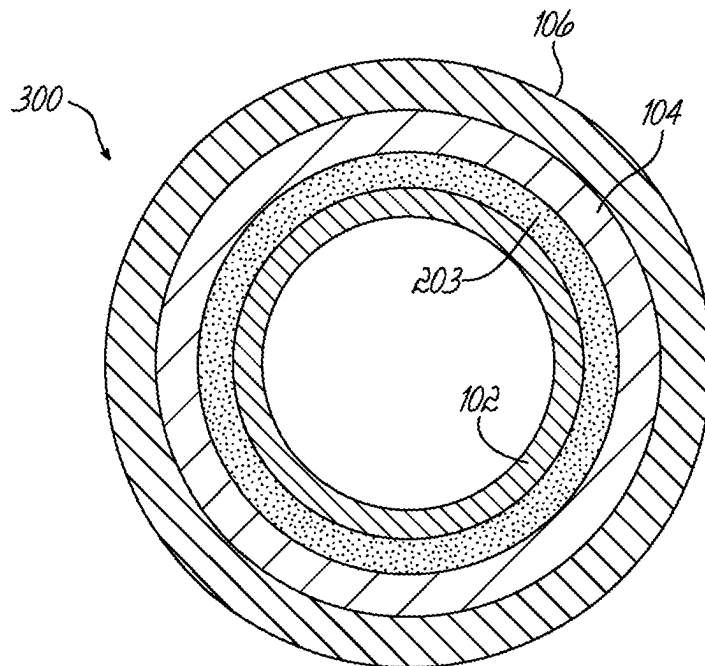


FIG. 3B

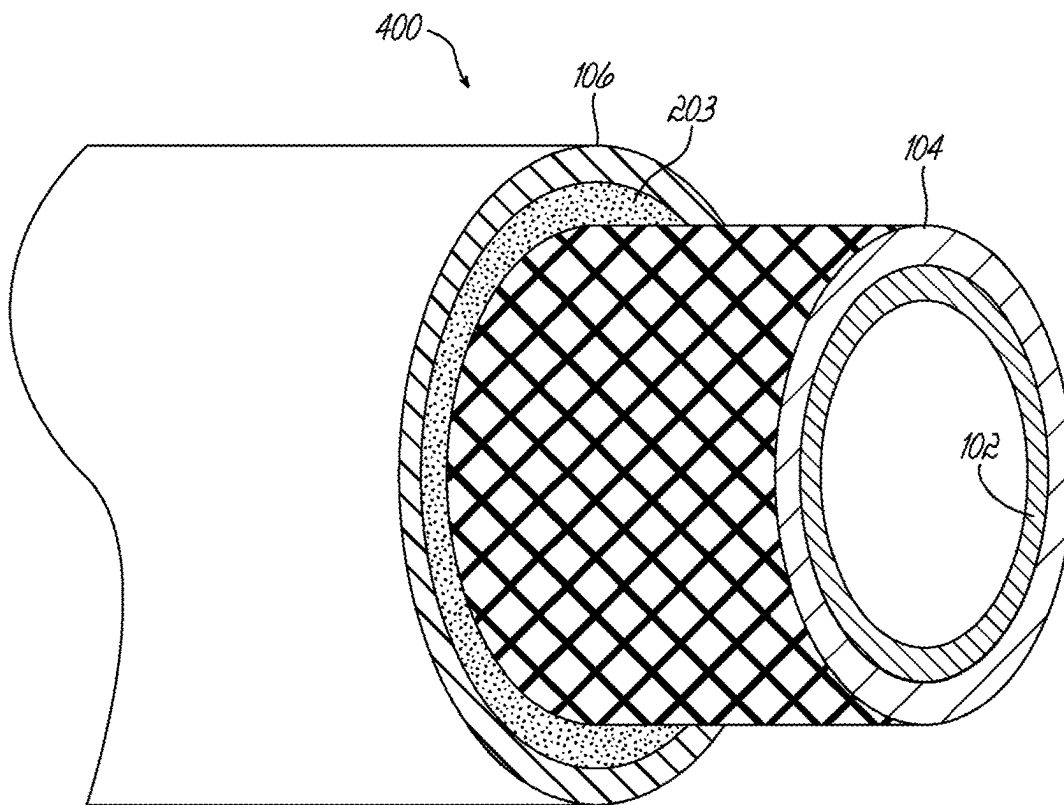


FIG. 4A

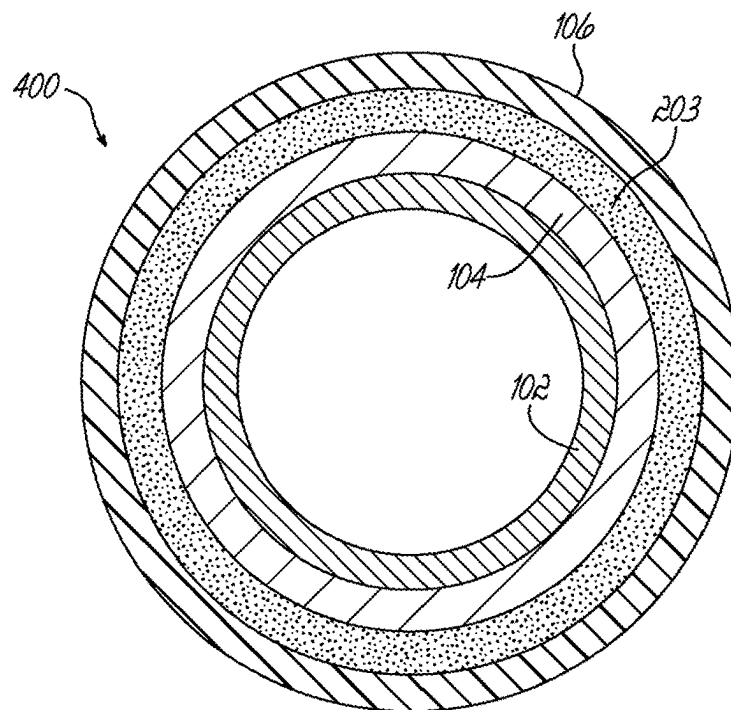


FIG. 4B

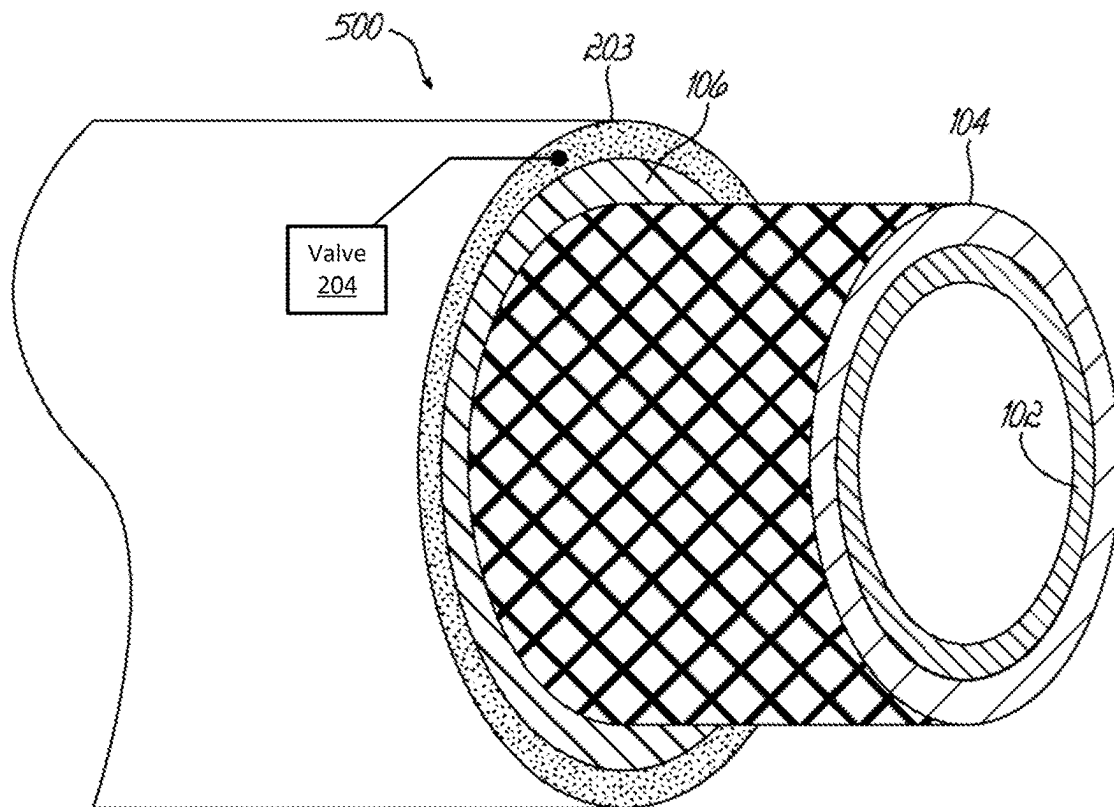


FIG. 5A

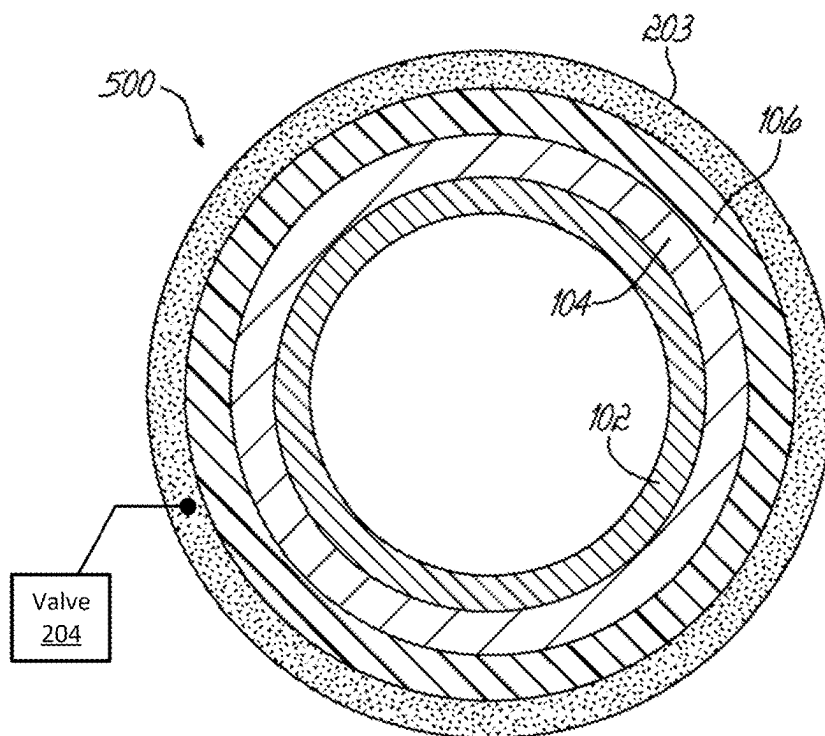


FIG. 5B

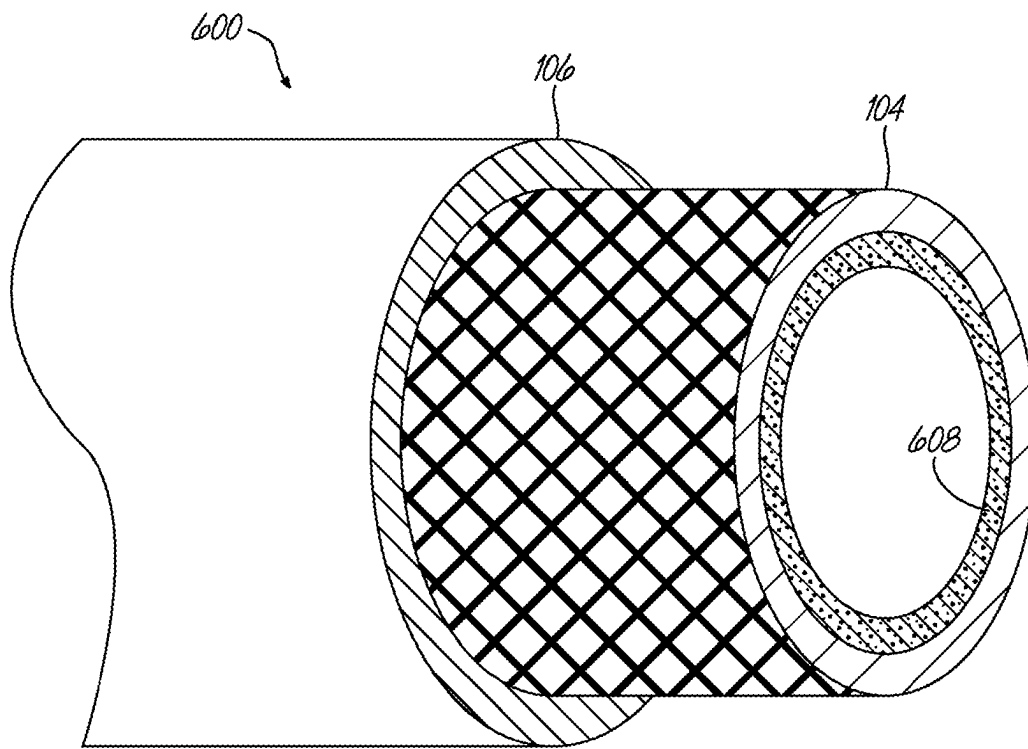


FIG. 6A

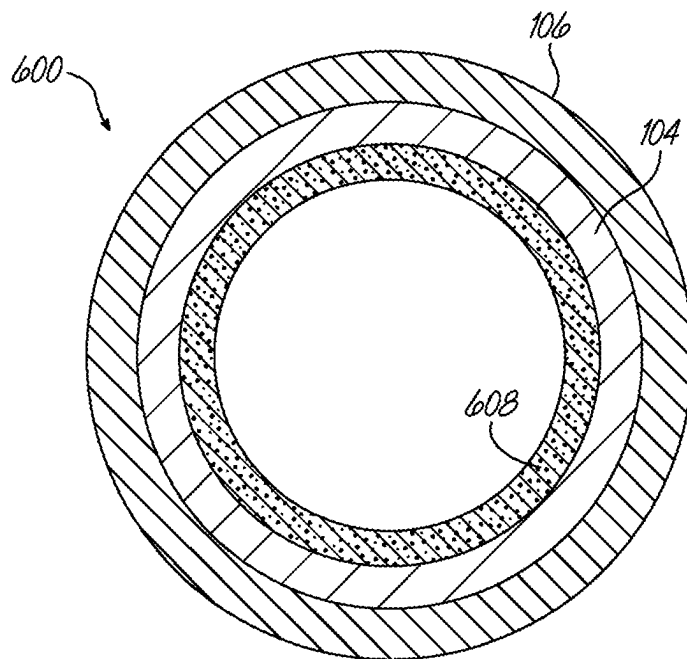


FIG. 6B

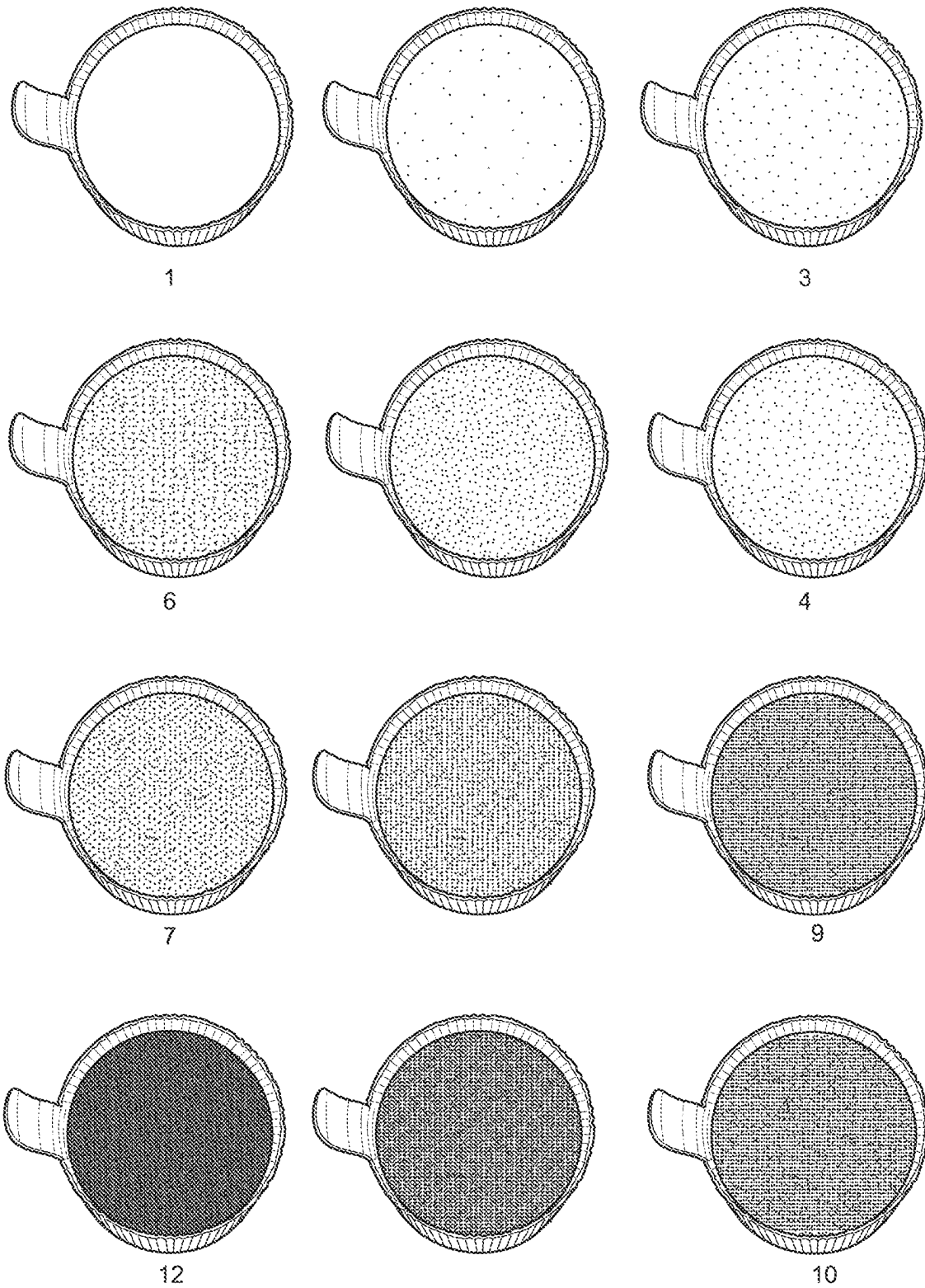


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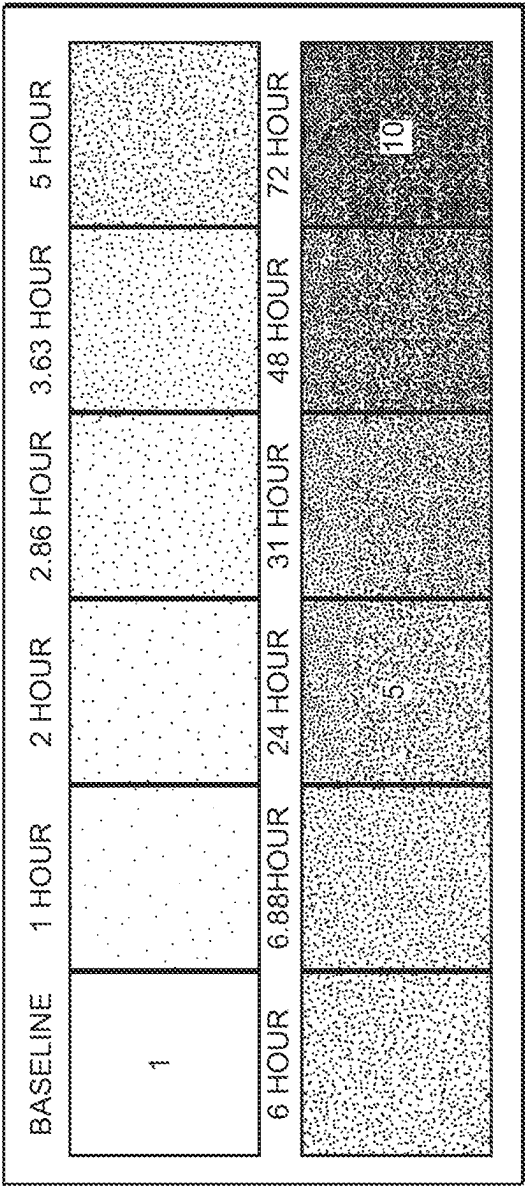


FIG. 8

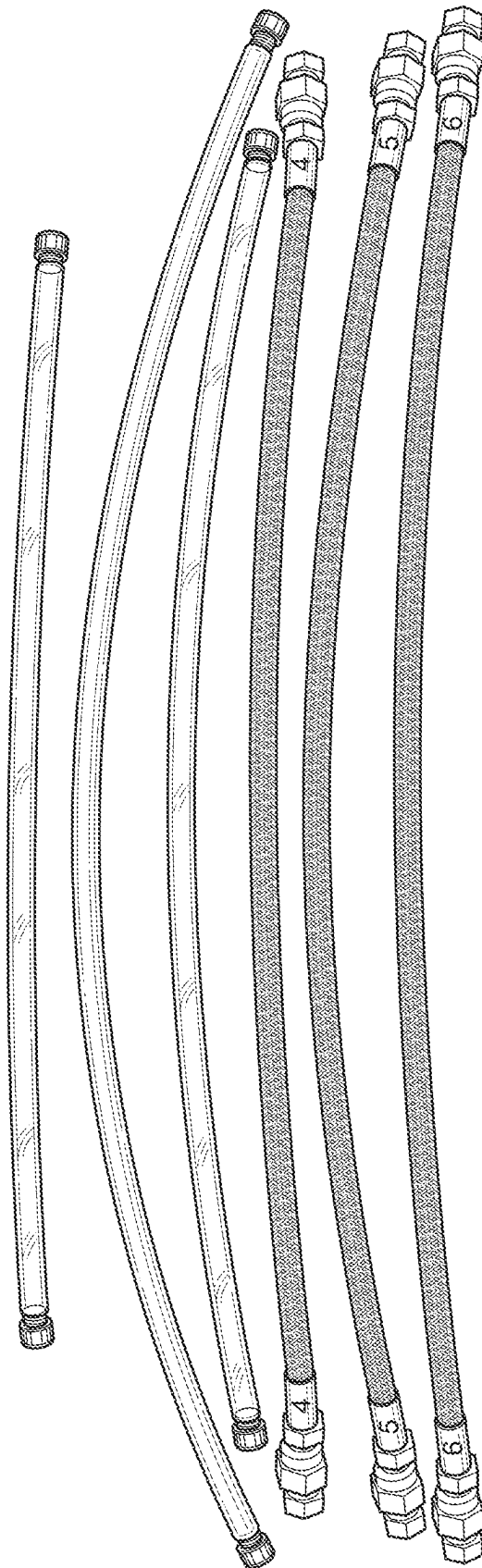
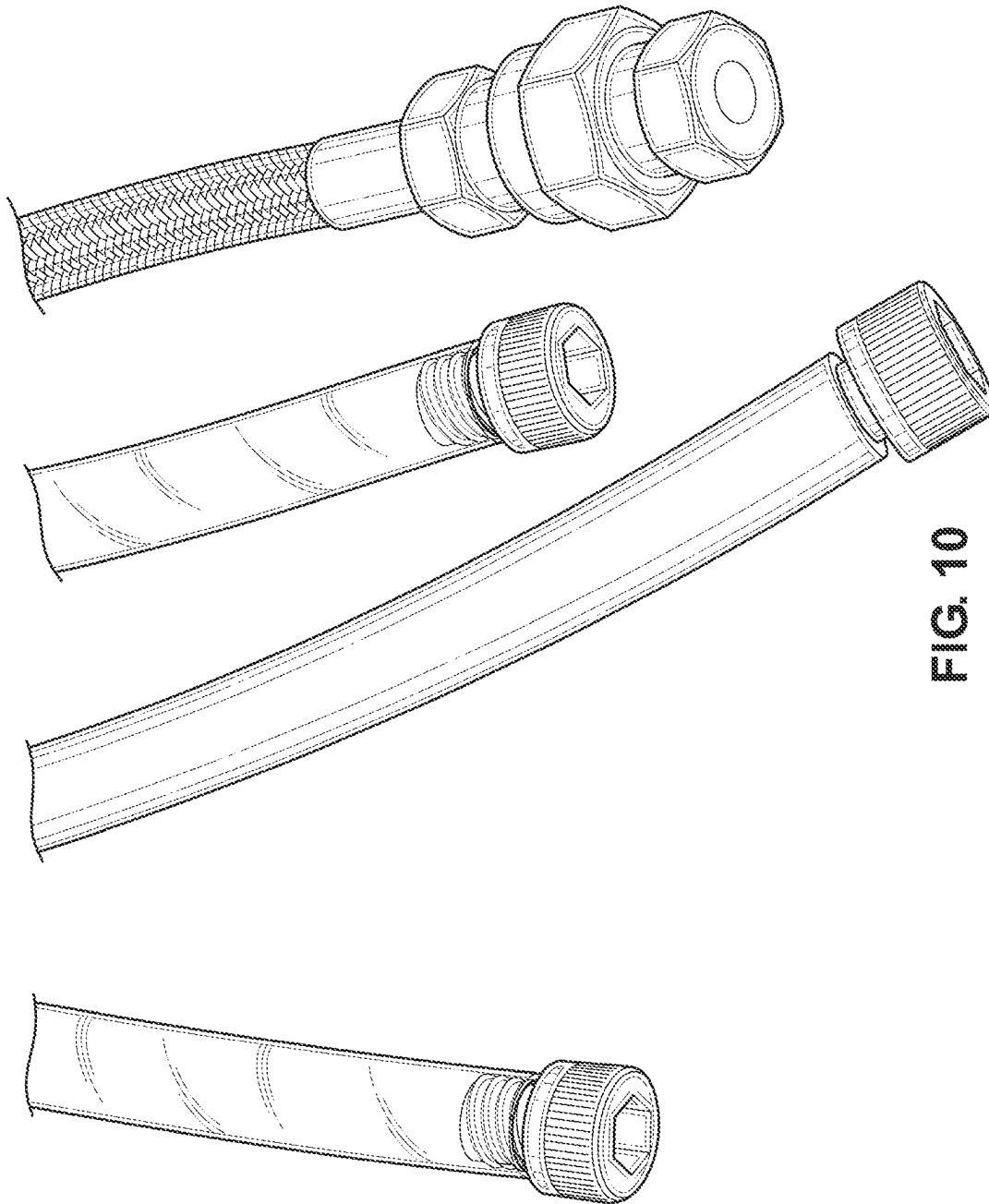


FIG. 9



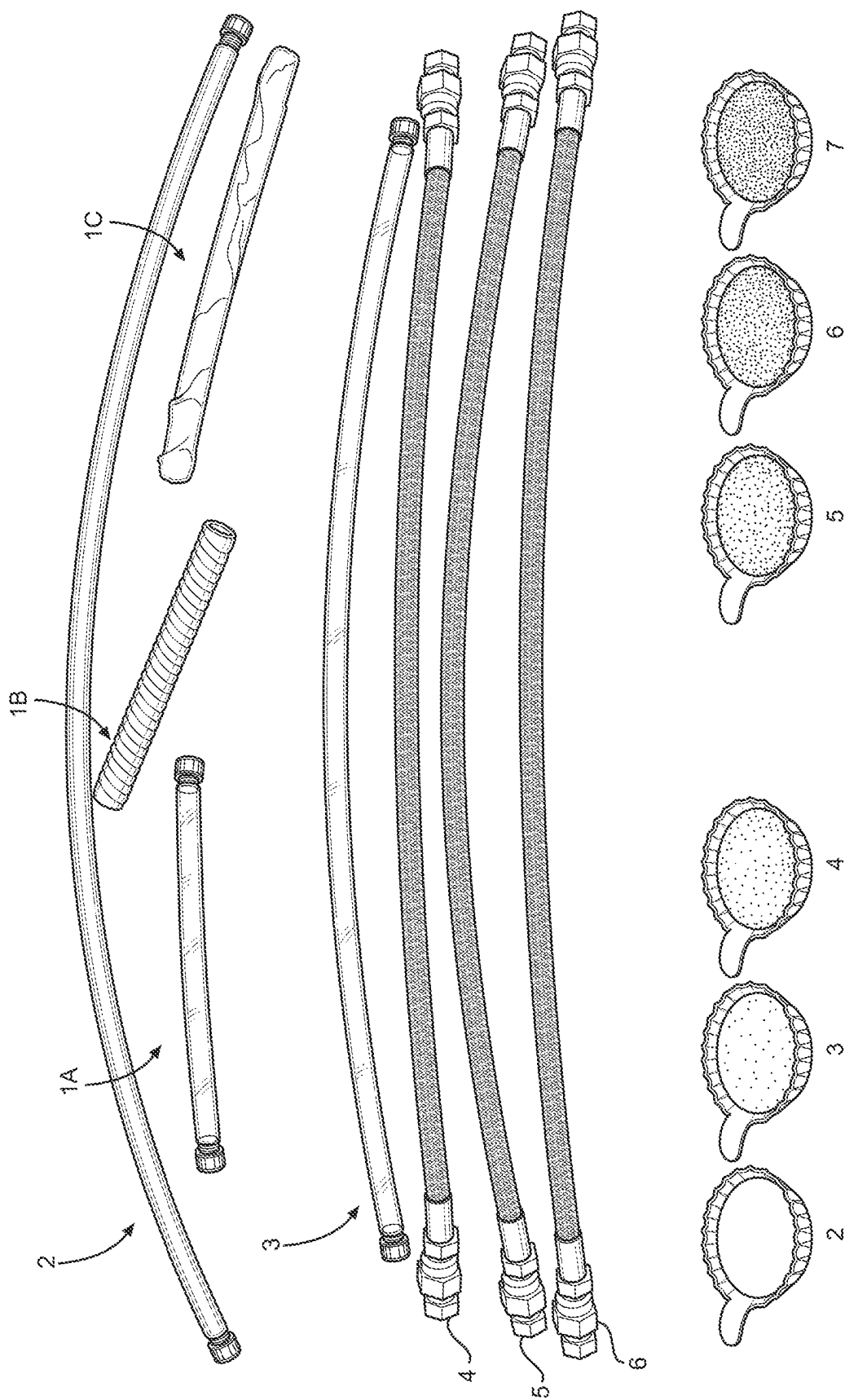


FIG. 11

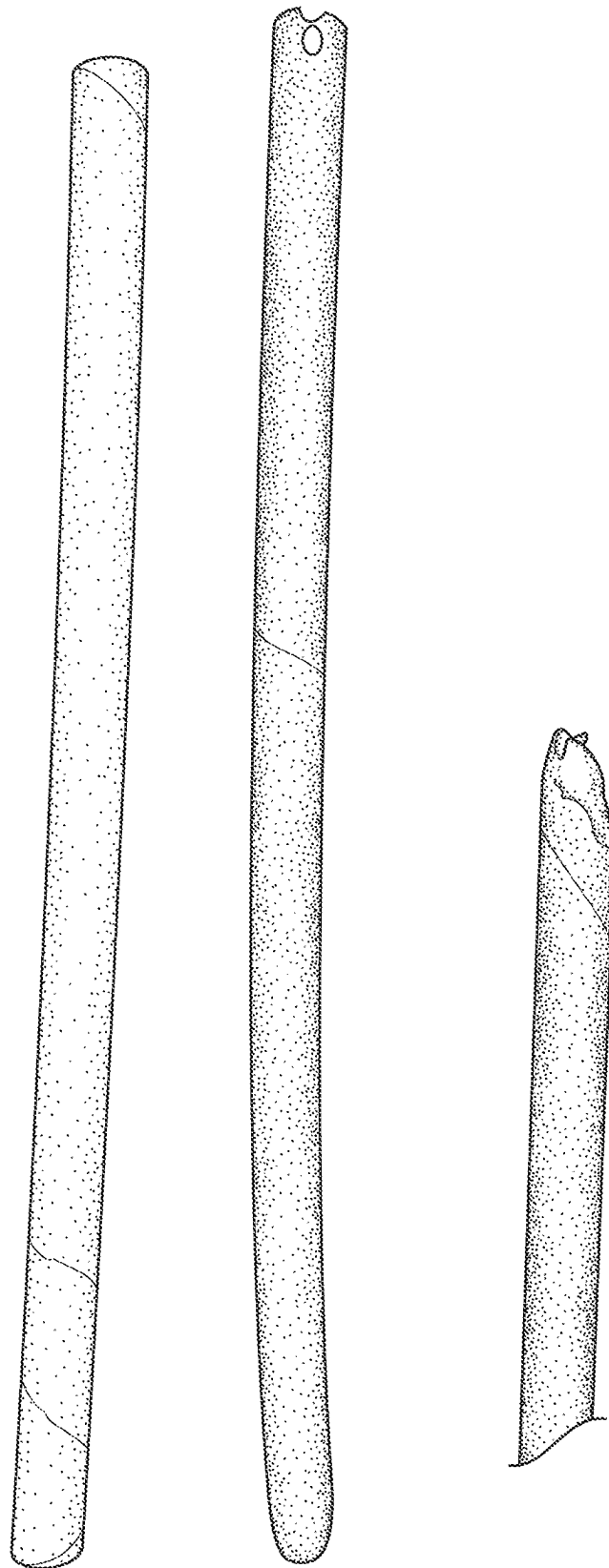


FIG. 12

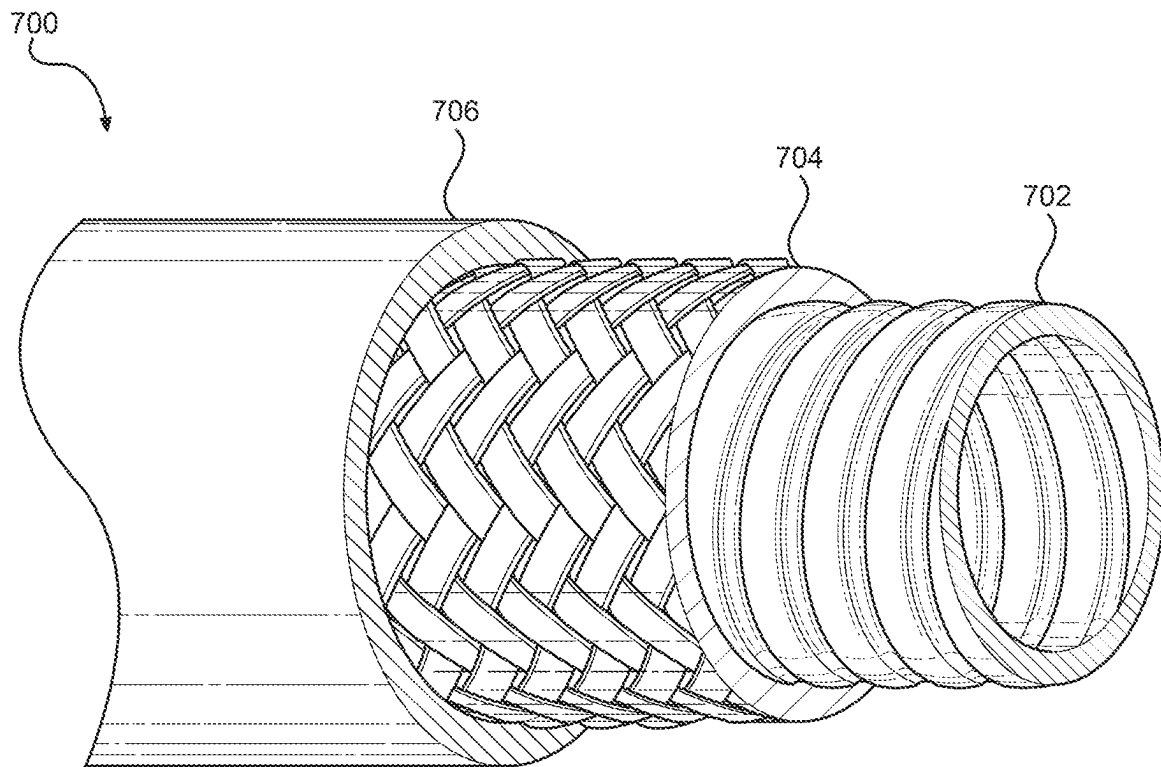


FIG. 13A

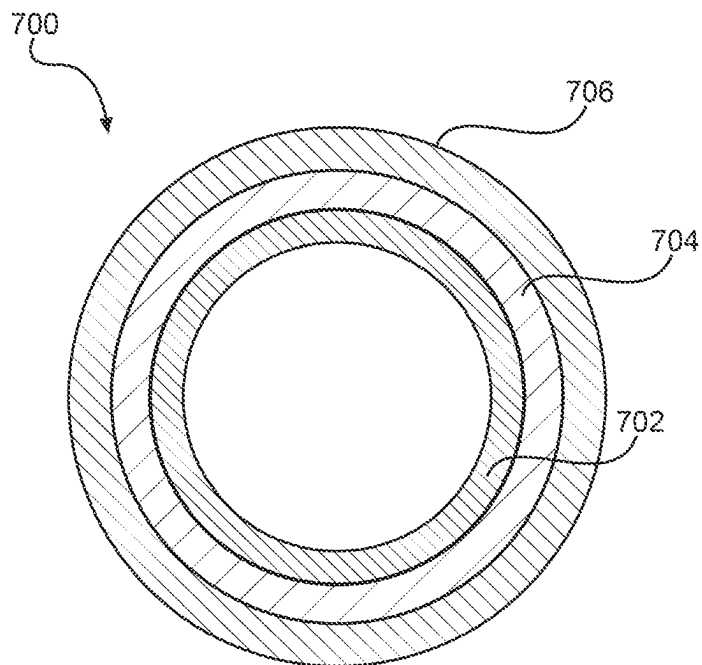


FIG. 13B

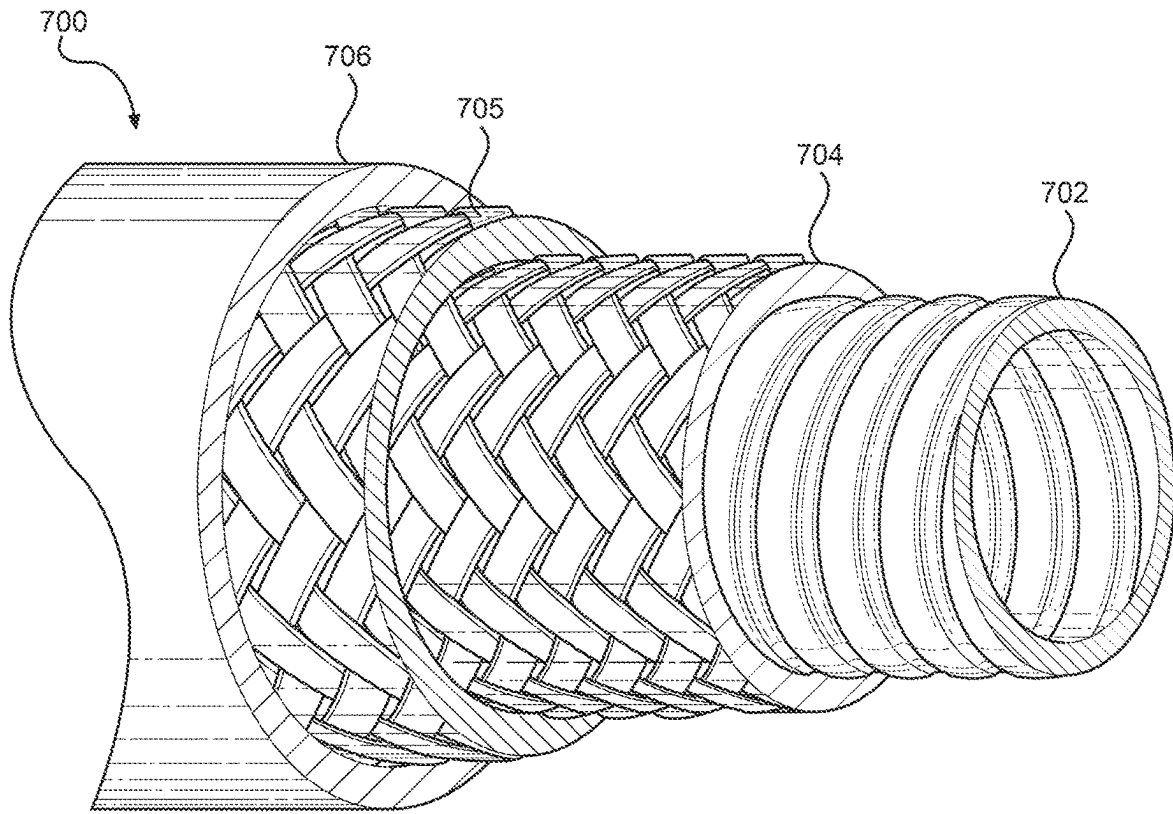


FIG. 14A

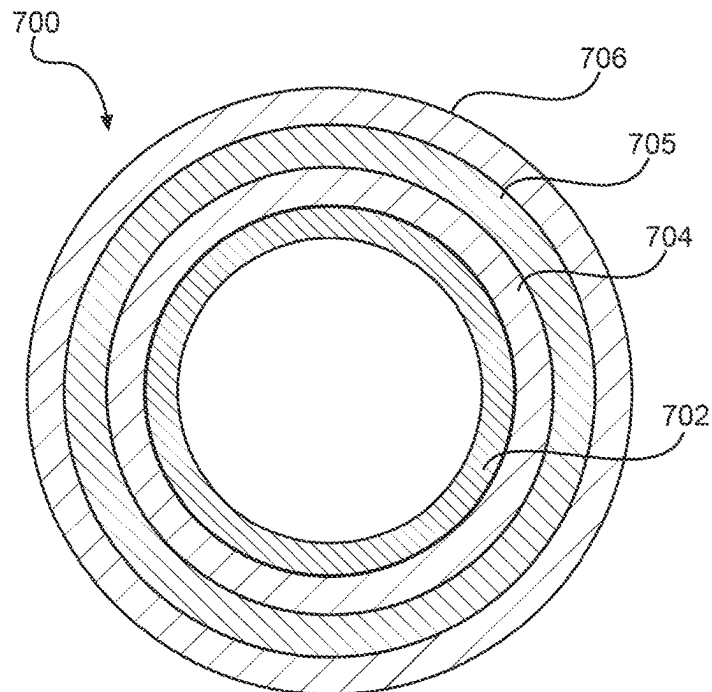


FIG. 14B

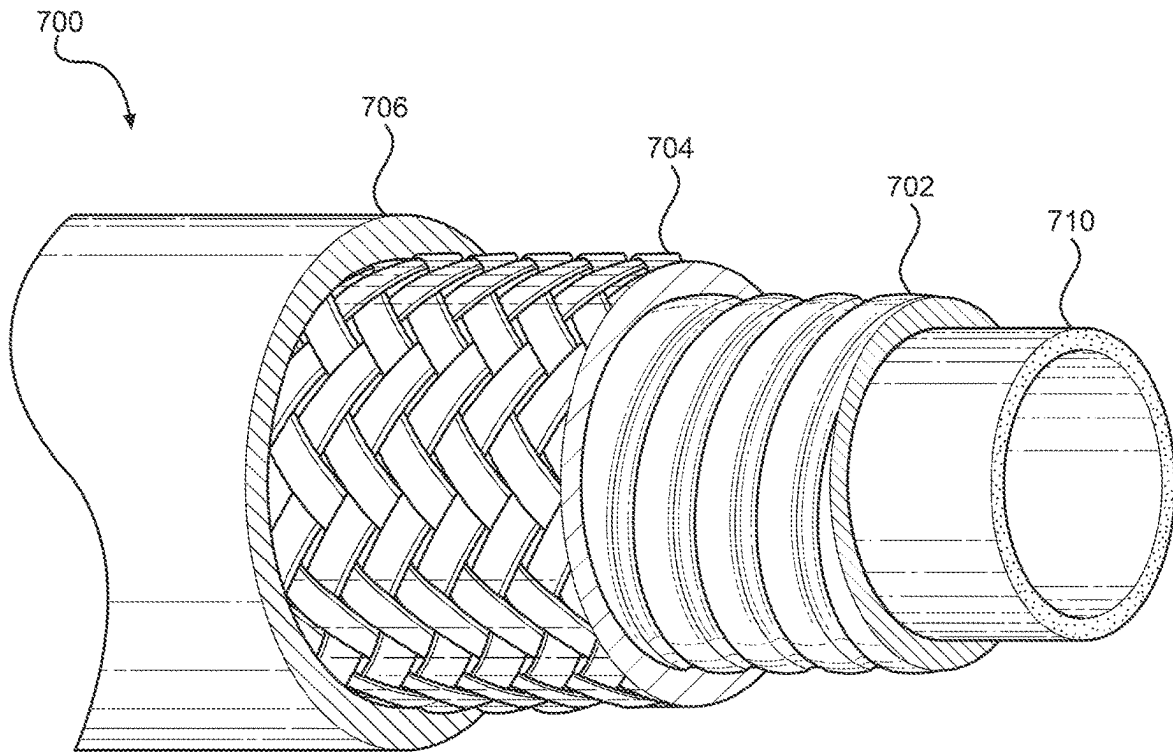


FIG. 15A

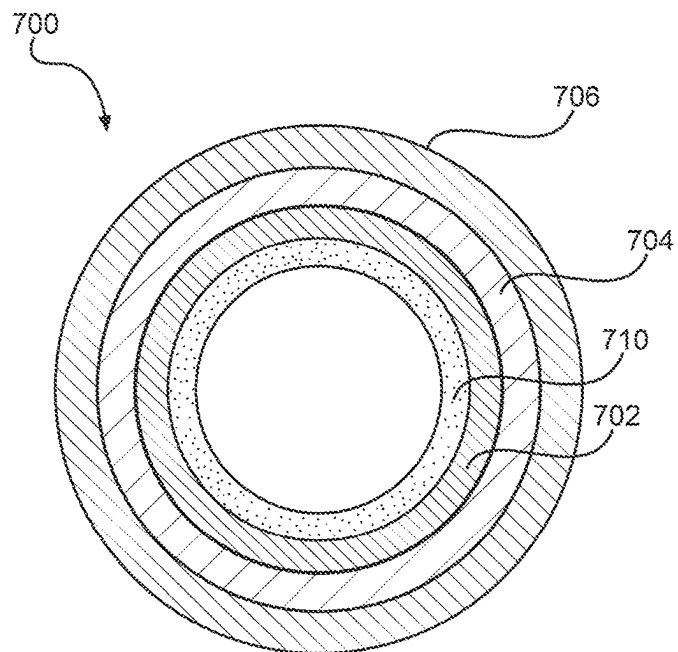


FIG. 15B

HOURS IN HOSE AT 350°F SETPOINT

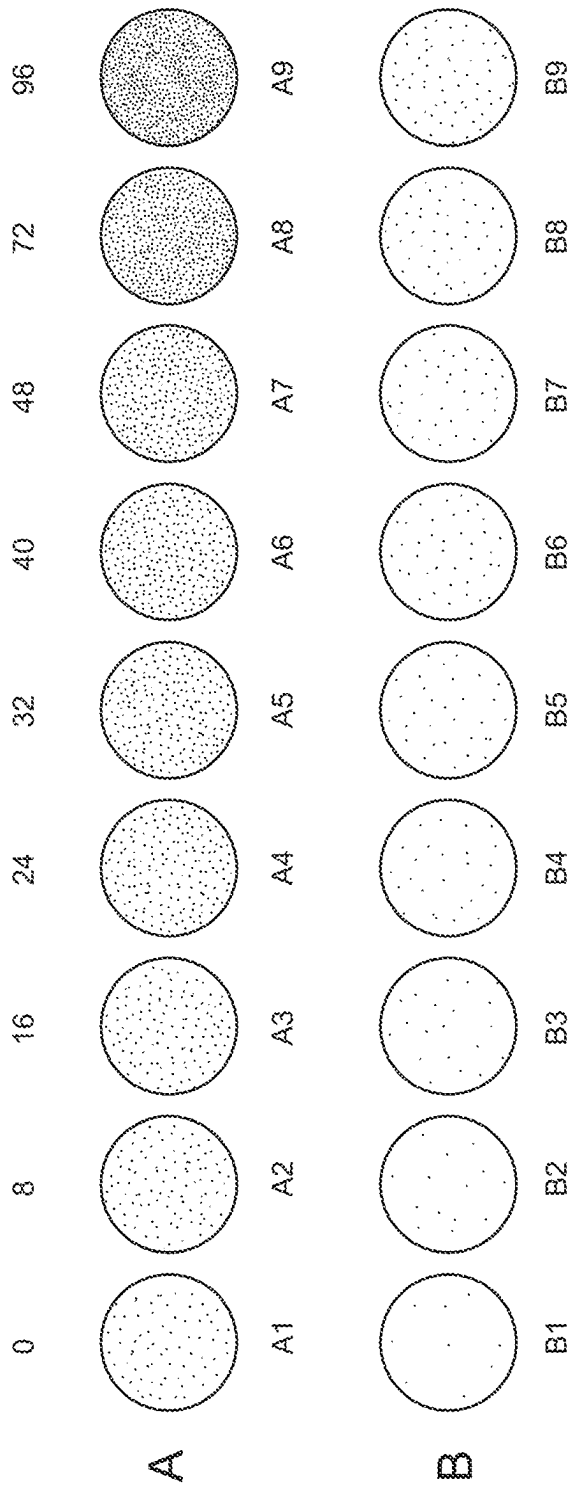
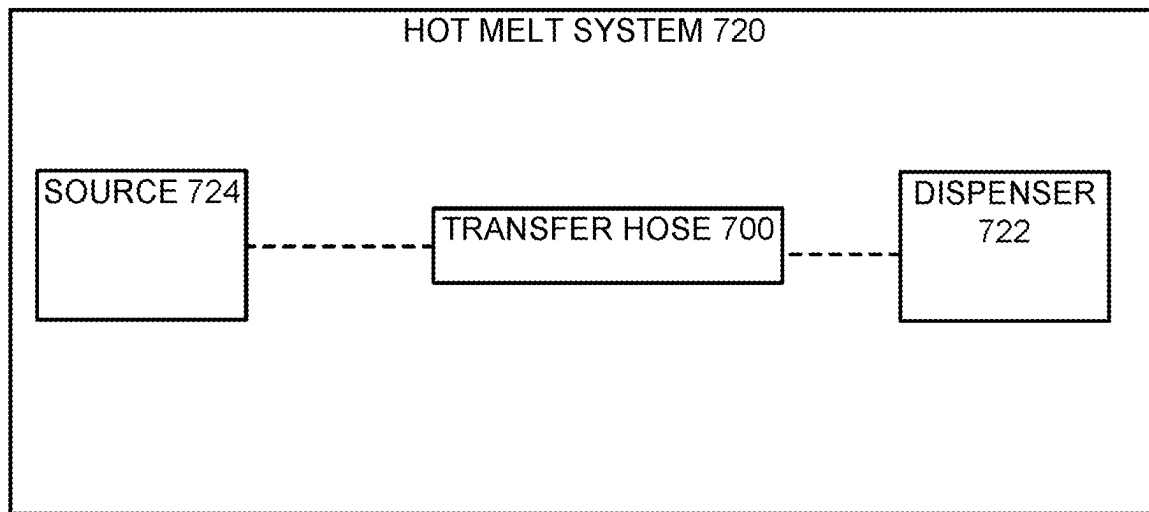
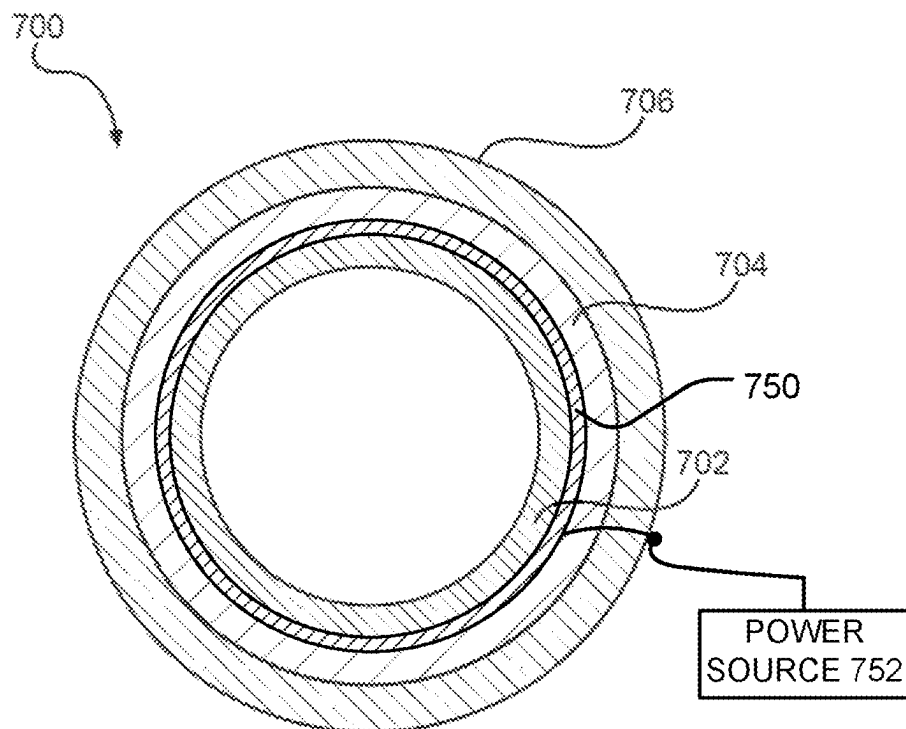
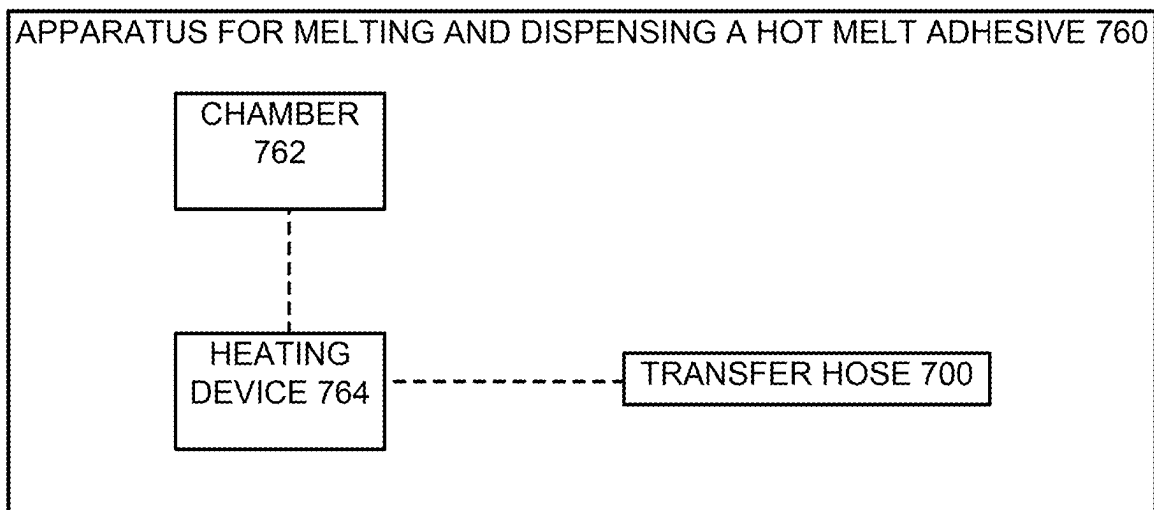


FIG. 16

**FIG. 17****FIG. 18**

**FIG. 19**

ADHESIVE TRANSFER HOSE HAVING A BARRIER LAYER AND METHOD OF USE

CROSS REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 15/632,371, filed Jun. 25, 2017, which claims the benefit of U.S. Provisional Patent Application No. 62/363,138, filed Jul. 15, 2016, the entire disclosures of both of which are hereby incorporated by reference as if set forth in their entirety herein.

TECHNICAL FIELD

The present disclosure is generally directed to adhesive transfer hoses, and more specifically, to adhesive transfer hoses having a barrier layer preventing or minimizing the ingress of air into a conduit of the hose.

BACKGROUND

Hot melt adhesives, which include conventional hot melt adhesives and reactive, moisture curing hot melt polyurethane adhesives ("hot melt PURs"), are routinely used in various applications where a stable surface-to-surface bond must be formed. Further, hot melt adhesives are used in securing a variety of both similar and dissimilar materials together in a mating relationship, such as wood, plastics, corrugated films, paper, carton stocks, metals, rigid polyvinylchlorides (PVCs), fabrics, leathers, and others. These adhesives are especially useful in applications where it is desirable to have the adhesive solidify rapidly after being melted and dispensed.

In conventional hot melt adhesives, typically a polymer, a tackifier, and a selection of other additives such as antioxidants, are mixed together to produce the adhesive. These materials tend to form bonds through their rapid solidification as they cool from a melted state and have the advantage of being relatively easy to apply. In hot melt PURs, an isocyanate terminated urethane polymer is produced through the polymerization of polyols and excess polyisocyanate compounds. Hot melt PURs cure (e.g., cross-link) in the presence of adventitious moisture.

Typically, a solid form of the hot melt adhesive comes in various shapes and sizes and is supplied to a melter that includes a heated tank and/or a heated grid to produce molten hot melt adhesive. Solid hot melt adhesive can also be supplied in drums or barrels in which the adhesive is melted by the use of a platen. After heating, the molten adhesive is pumped through a heated hose that maintains the molten material at the required application temperature, to an applicator or dispenser, which is sometimes referred to as a dispensing "gun" or a gun module, comprising a valve and a nozzle. An exemplary heated hose is described in commonly owned U.S. Pat. No. 6,738,566, which is incorporated herein by reference in its entirety.

However, a common problem with heated hoses is the resulting discoloration of the molten hot melt adhesive. Such discoloration may negatively affect the "pot life" of the hot melt adhesive, since the discoloration can indicate that the hot melt adhesive has degraded. "Pot life" as used herein is the maximum time at the system temperature before the adhesive starts to degrade, thus resulting in increased viscosity, charring and/or gelling. This could be especially problematic in systems requiring relatively low flow rates. In some applications, the "pot life" of the molten adhesive

can shorten if it stays within a heated hose for too long. Shortening the "pot life" of a thermoplastic adhesive may result in operational problems, such as filter clogging, and may further require cleaning of the hose after charring has occurred.

Char is adhesive that has been blackened or burned, and can result from a variety of reasons, such as heating hot melt adhesive for too long and/or heating it at too-high of a temperature. Additionally, it has been observed that the introduction of oxygen into the hose is a main cause of charring of the hot melt adhesive. Although hot melt adhesives can be protected by certain additives, like antioxidants, hot melt adhesives should not be kept in the molten state for an extended period of time since they can break down. Thus, the effects of heat, time and oxidation begin to break down the adhesive. For instance, the adhesive's polymer chains form active sites that can combine to form gels which stick to the walls of hoses and crevices in melt tanks, forming an anchor that inhibits effective flow of the hot melt adhesive through the system. Moreover, char can harden and break off into pieces that clog filters and spray nozzles.

A major problem with char is that once it gets into a hot melt system, it is very difficult and sometimes impossible to flush out. Once char forms it can cause ongoing product quality problems, extensive maintenance issues and work stoppages. In some cases, the entire hot melt system may need to be taken apart and the components must be burned out in a burnout oven to completely remove the char. This process is very time-consuming and expensive. Therefore, it is desirable to prevent or reduce char formation before it can become a problem in the system.

The inventors of the present disclosure have found that one way to prevent charring of the adhesive is by eliminating or minimizing ingress of oxygen within the hose. However, preventing or reducing the introduction of oxygen into conventional adhesive transfer hoses can pose substantial problems since typical hoses that are long and have a small diameter provide a large transfer area for a small amount of adhesive. Thus, the ingress of oxygen into the transfer hose is more likely than the ingress of oxygen into the adhesive melting tank. For example, the adhesive can sit in a heated hose for a long duration of time and experience minimal discoloration as long as oxygen is prevented from entering the hose during transfer of the adhesive.

Moreover, conventional hot melt hoses do not take into account the impact of the partial pressure of oxygen on diffusion through a typical hose core. The antioxidant and degradation processes consume oxygen within the hose, creating a driving force for diffusion from the atmosphere, despite the fact that the hose may be hydraulically pressurized to several hundred psi. Moreover, hoses that are used for high temperature applications of 500° F. and above, such as for polymer processing, are well beyond those required to sufficiently melt the solid form of adhesive to a molten flowable state.

Thus, discoloration and associated degradation of the hot melt adhesive are the result of adventitious oxygen that penetrates into and/or through the layers of the hot melt adhesive transfer hose and reacts with the hot melt adhesive under the molten temperature conditions. Accordingly, there is a need for a hot melt adhesive transfer hose having an impermeable barrier layer which serves to prevent or minimize the ingress of oxygen into the conduit of the hose that transfers the molten hot melt adhesive, thereby eliminating or at least greatly reducing the discoloration and corresponding charring of the molten hot melt adhesive during its residence time in the heated hose.

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SUMMARY

A multi-layered hot melt adhesive transfer hose is provided. The transfer hose comprises a conduit for transporting heated liquid hot melt adhesive, an impermeable barrier layer configured to prevent the ingress of oxygen into a conduit of the hose that transports a molten hot melt adhesive; and at least one structural layer overlaying an exterior surface of the barrier layer and configured to withstand a high fluid pressure. More particularly, the barrier layer is configured to prevent ingress of oxygen from passing through the conduit and into the hot melt adhesive. A heater may also be provided for maintaining the hot melt adhesive at a set point temperature, wherein the heater is adapted to be electrically coupled to a power source controlled to maintain the set point to about 450° F. or less for an extended period of time.

The barrier layer and the structural layer each comprise a thermally stable material. Further, the barrier layer prevents the hot melt adhesive from charring when the adhesive is heated to about 450° F. or less for an extended period of time. The barrier layer is a flexible metal tube that can include a plurality of seamless corrugations. Alternatively, the barrier layer may be a metal tape, such as aluminum tape.

The at least one structural layer can include a braided jacket, which may be stainless steel. The flexible metal tube may also be stainless steel. Further, the at least one structural layer can include two overlapping braided jackets, and an outer covering layer has a distinct multi-layered structure overlaying an exterior surface of the at least one structural layer.

The outer covering layer can include a heating wire sublayer, an insulation sublayer covering the heating wire sublayer, and a protective sublayer covering the insulation sublayer. It should be appreciated that the outer covering layer can alternatively include a heating tape sublayer.

Further, the barrier layer may be a metallic coating applied to the inner tube. It should be appreciated that the flexible metal tube may include a liner disposed within. The liner comprises a heat stable polymeric material and is configured to improve flow of the molten hot melt adhesive through the flexible metal tube. Moreover, the liner includes a smooth interior surface that facilitates fluid flow and prevents material incompatibility issues between the molten hot melt adhesive and the flexible metal tube.

The barrier layer unexpectedly prevents the hot melt adhesive from discoloring and charring when the adhesive is heated at a temperature of about or greater than 250° F. up to and including about 450° F. for an extended period of time.

Further, a method of transporting hot melt adhesive is disclosed, including the step of transporting the hot melt adhesive at a temperature at or below about 450° F. through a multi-layered transfer hose comprising a flexible metal tube forming an impermeable barrier layer configured to prevent the ingress of oxygen into a conduit of the hose. It should be appreciated that the flexible metal tube is heated to a temperature at or below about 450° F. Alternatively, the flexible metal tube can transport hot melt adhesive at a temperature at or below about 400° F. Furthermore, the flexible metal tube can transport hot melt adhesive at a temperature at or below about 350° F.

An unexpected effect of the barrier layer is that no significant discoloration and charring of the hot melt adhesive occurs when the adhesive remained inside the hose for at least twenty-four hours, at least forty-eight hours, seventy-two hours, or ninety-six hours.

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In accordance with another aspect of the present disclosure, a multi-layered hot melt adhesive transfer hose is provided, wherein at least one of the layers is an oxygen barrier layer that prevents or minimizes the ingress of oxygen into a conduit of the hose that transports a molten hot melt adhesive.

In accordance with another aspect of the present disclosure, an apparatus for melting and dispensing a hot melt adhesive is provided. The apparatus comprises a chamber for receiving a solid form of the hot melt adhesive; a heating device coupled to the chamber and configured to receive the hot melt adhesive for liquefying the solid form of the hot melt adhesive; and the multi-layered hot melt adhesive transfer hose fluidly coupled to the heating device.

Further, a method is provided for dispensing hot melt adhesive, such as a packaging grade hot melt adhesive. The method includes the steps of melting hot melt adhesive; transporting the melted hot melt adhesive through a hose to a dispenser; heating the melted hot melt adhesive in the hose at a set point temperature of about 450° F. or less; preventing, with a barrier layer, the transfer of oxygen into the adhesive within the hose; and dispensing the hot melt adhesive onto a substrate.

This method for dispensing hot melt adhesive produces no significant discoloration and charring of the adhesive when it remains inside the hose for at least 24 hours. Moreover, no significant discoloration and charring of the hot melt adhesive occur when the adhesive remains inside the hose for at least 48 hours to 96 hours.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a perspective schematic view of a conventional hot melt adhesive transfer hose.

FIG. 1B is an end view of the hot melt adhesive transfer hose shown in FIG. 1A.

FIG. 2A is a perspective schematic view of a hot melt adhesive transfer hose, in accordance with an embodiment of the present invention.

FIG. 2B is an end view of the hot melt adhesive transfer hose shown in FIG. 2A.

FIG. 3A is a perspective schematic view of a hot melt adhesive transfer hose, in accordance with another embodiment of the present invention.

FIG. 3B is an end view of the hot melt adhesive transfer hose shown in FIG. 3A.

FIG. 4A is a perspective schematic view of a hot melt adhesive transfer hose, in accordance with yet another embodiment of the present invention.

FIG. 4B is an end view of the hot melt adhesive transfer hose shown in FIG. 4A.

FIG. 5A is a perspective schematic view of a hot melt adhesive transfer hose, in accordance with yet another embodiment of the present invention.

FIG. 5B is an end view of the hot melt adhesive transfer hose shown in FIG. 5A.

FIG. 6A is a perspective schematic view of a hot melt adhesive transfer hose, in accordance with an embodiment of the present invention.

FIG. 6B is an end view of the hot melt adhesive transfer hose shown in FIG. 6A.

FIG. 7 illustrates pan samples of Henkel 314C hot melt adhesive that have been heated from 0 to about 95 hours at 350° F. in a laboratory oven showing various states of discoloration.

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FIG. 8 shows an illustrative qualitative discoloration grading scale from 1 to 10, which is based on the samples shown in FIG. 7.

FIG. 9 illustrates various exemplary test hoses.

FIG. 10 illustrates the end caps of the exemplary test hoses shown in FIG. 9.

FIG. 11 illustrates placement of the testing samples reported below in Table 1 before beginning an experiment described herein.

FIG. 12 illustrates samples 1b, 1a, and 1c (left to right) after the completion of the experiment described herein.

FIG. 13A is a perspective schematic view of a hot melt adhesive transfer hose in accordance with the present disclosure.

FIG. 13B is an end view of the hot melt adhesive transfer hose shown in FIG. 13A.

FIG. 14A is a perspective schematic view of a hot melt adhesive transfer hose in accordance with another aspect of the present disclosure.

FIG. 14B is an end view of the hot melt adhesive transfer hose shown in FIG. 14A.

FIG. 15A is a perspective schematic view of a hot melt adhesive transfer hose in accordance with another aspect of the present disclosure.

FIG. 15B is an end view of the hot melt adhesive transfer hose shown in FIG. 15A.

FIG. 16 is an illustration comparing the amount of char formed in hot melt adhesive over time from using a competitor's transfer hose compared to the transfer hose of the present disclosure.

FIG. 17 is a schematic illustration of a hot melt system that includes a dispenser operatively coupled to the hot melt adhesive transfer hose of FIG. 13A.

FIG. 18 is a schematic illustration of an end view of a hot melt adhesive transfer hose that includes a heater that is coupled to a power source.

FIG. 19 is a schematic illustration of an apparatus for melting and dispensing a hot melt adhesive, including the hot melt adhesive transfer hose of FIG. 13A coupled to a heating device that is coupled to a chamber.

It should be noted that the figures are not necessarily drawn to scale, but instead are drawn to provide a better understanding of the components thereof, and are not intended to be limiting in scope, but rather to provide exemplary illustrations. Further, implementations of the present disclosure are described with reference to the drawings, in which like reference numerals refer to like parts throughout.

DETAILED DESCRIPTION

As noted above, discoloration of the hot melt adhesive occurring in a conventional hot melt adhesive transfer hose may negatively affect the "pot life" of the hot melt adhesive by subsequently resulting in increased viscosity, charring, or gelling. It was observed that such discoloration could be derived from adventitious oxygen that had penetrated into and/or through the layers of the hot melt adhesive transfer hose and reacted with the hot melt adhesive under the molten temperature conditions.

Thus, a multi-layered hot melt adhesive transfer hose that includes a barrier layer is disclosed. The barrier layer serves to prevent or minimize the ingress of oxygen into the conduit of the hose that transfers the molten hot melt adhesive, and thereby reduces the discoloration of the molten hot melt

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adhesive. Further, the inhibition or reduction in oxygen ingress may also preserve the expected "pot life" of the hot melt adhesive.

The oxygen barrier layer may be a distinct layer of the hose, or a composite or mixture of a heat stable polymer and an inorganic additive that functions as the inner tube, as will be explained in more detail below. A hot melt adhesive apparatus that incorporates one or more of the multi-layered hot melt adhesive transfer hoses, as well as a method for transferring hot melt adhesive and making the hot melt adhesive transfer hose, are also described.

In one aspect of the disclosure, the barrier layer is impermeable and thus prevents oxygen from diffusing into the conduit of the hose. In another aspect of the disclosure, the oxygen barrier layer provides a level of oxygen permeability to the hot melt adhesive transfer hose that is low enough to reduce the discoloration as compared to one void of the oxygen barrier layer. For example, the oxygen permeability of the hose with the oxygen barrier layer may be reduced by a factor of about 10 or about 100 or about 1,000 or more.

Referring to FIGS. 1A and 1B, a simplified multi-layered hot melt adhesive transfer hose 100 of the prior art is shown having an inner tube 102 for conveying the molten hot melt adhesive, a structural layer 104 for strength and protection, and an outer covering 106. The inner tube 102 forms the operative core of the hose 100 through which the molten hot melt adhesive actually flows.

The inner tube 102 is made from a polymeric material capable of withstanding relatively high temperatures such as polytetrafluoroethylene (PTFE). Since PTFE or other similar high melting temperature polymers are typically unable to withstand the high fluid pressure used to transfer the molten hot melt adhesive, the inner tube 102 needs to be reinforced by a reinforcing layer, or structural layer 104.

The structural layer 104, which is disposed about the exterior of the inner tube 102 and serves to provide strength and protection to the inner tube 102, may comprise a braided jacket of thermally stable material. As noted above, hot melt adhesives are heated to set point temperatures sufficient to melt the solid form to a molten, flowable state, which are generally in a range from about 100° C. (about 212° F.) to about 230° C. (about 450° F.). Additionally, to facilitate flow of the molten hot melt adhesive, the hot melt adhesive transfer hose may experience pressures up to about 1500 psi (about 10.3 MPa) operating pressures. Accordingly, the structural layer 104 serves to provide desired physical integrity of the hot melt adhesive transfer hose.

Polymeric Barrier Layer

Thus, the present disclosure provides a multi-layered hot melt adhesive transfer hose 200 that is configured to prevent and/or reduce gasses, such as oxygen, from penetrating the hose and contacting the hot melt adhesive therein. As shown in FIGS. 2A and 2B, an oxygen barrier layer 203 of the hot melt adhesive transfer hose 200 may be a distinct layer that is in contact with an inner surface of the inner tube 102, which is in turn covered by the structural layer 104 and the outer covering 106. Alternatively, as shown in FIGS. 3A and 3B, the oxygen barrier layer 203 may be a distinct layer that is in contact with an outer surface of the inner tube 102 or an inner surface of the structural layer 104, which in turn is covered by the outer covering 106. As shown in FIGS. 4A and 4B, the oxygen barrier layer 203 may be a distinct layer that is circumferentially outside of the inner tube 102 and in contact with an outer surface of the structural layer 104 or an inner surface of the outer covering 106. As shown in FIGS. 5A and 5B, the oxygen barrier layer 203 may encom-

pass the inner tube **102**, the structural layer **104**, and the outer covering **106**, and is formed as a distinct layer that is in contact with an outer surface of the outer covering **106**.

Referring to FIGS. 2-5, the inner tube **102** forms the operative core of the hose **200** through which the molten hot melt adhesive actually flows. The oxygen barrier layer **203** may include a polymeric material, a metallic material, or a combination thereof. The selection of the material may be dependent upon the location of the oxygen barrier layer in the hot melt adhesive transfer hose. For example, the inner tube **102**, the structural layer **104**, and/or the heat tape sublayer are in what may be referred to as a "hot zone," which is near or above the temperature of the molten hot melt adhesive. Alternatively, the outer surface of an insulation or the protective sublayers of the outer layer **106** are radially disposed from and are outside the hot zone, and thus are at a lower temperature. Accordingly, materials having melting temperatures above the desired working temperature of the molten hot melt adhesive may be used in constructing the oxygen barrier layer within or outside of the hot zone.

When the oxygen barrier layer **203** is within the hot zone, the melting point of the material constructing the oxygen barrier layer **203** should have a melting point sufficiently above the desired working temperature of the molten hot melt adhesive. For instance, the melting point of the material constructing the oxygen barrier layer **203** within the hot zone is preferably above the desired working temperature of the molten hot melt adhesive by at least about 50° F. or more, or about 100° F. or more, or 200° F. or more. Examples of such materials having a sufficiently high melting point include, but are not limited to, metallic materials such as metal foils or metal coatings. Non-limiting examples of the metallic materials include an aluminum foil backed tape, or a metal or metallic coating applied by sputtering, chemical vapor deposition (CVD), plasma-enhanced chemical vapor deposition (PECVD), or atomic layer deposition (ALD). The surface of the layer onto which the metal or metallic coating is applied may be modified to improve metal adhesion, such as that described in U.S. Pat. No. 6,420,041, which is incorporated by reference herein in its entirety. The thicknesses of the metal materials useful for forming the oxygen barrier layer may vary depending on the desired level of reduction in oxygen permeability.

With respect to the oxygen barrier layer **203** comprising polymeric materials, either alone or in combination with metallic materials, the melting point of the polymeric materials may render the placement of the oxygen barrier layer **203** according to the placement depicted in one of FIGS. 2-5 more preferable over another. A list of exemplary polymeric materials having desirable oxygen permeability values is shown in Table 1:

TABLE 1

Gas Permeability of Select Polymers		
Polymer	Abbreviation, Product Name, etc.	Oxygen permeability (cc/cm ² /mm/sec/cmHg × 10 ¹⁰)
Polytetrafluoroethylene	PTFE, Teflon™	—
Perfluoro ethylene-propylene copolymer	FEP	59
Polyvinylidene chloride	PVDC, Saran™	0.05
Polyvinyl Fluoride	PVF, Tedlar®	0.2
Polyethylene terephthalate	PET	0.3
Polychlorotrifluoroethylene	PCTFE	0.25-5.4

TABLE 1-continued

Gas Permeability of Select Polymers		
Polymer	Abbreviation, Product Name, etc.	Oxygen permeability (cc/cm ² /mm/sec/cmHg × 10 ¹⁰)
Phenol resin	PF, Bakelite®	—
Polyamide	PA, Nylon	0.38
Epoxy resin	EP	0.49-16
Polystyrene	PS-MAN	1.6
methacrylonitrile		
Polyacetal (homopolymer)	POM	0.38
Polyvinyl chloride	PCV	1.2-6
Styrene-acrylonitrile	—	3.4
Cellulose acetate	CA	4.0-7.8

*** Table obtained from <http://junkosha.co.jp/technical/tec8.html>, which was adapted from Teflon® Handbook of Dupont-Mistubishi Fluorochemicals Co. Ltd.

In addition, the polymeric material may comprise a urethane, such as a thermoplastic polyether-urethane (TPEU) or thermoplastic polyester-polyurethane elastomer, as disclosed in U.S. Pat. No. 9,192,754.

To further reduce the oxygen permeability of the polymeric material, the polymeric material may also be combined with an inorganic additive, such as clays, silicates and silicas, pillared materials, metal salts, nanoplatelets, or mixtures thereof, such as those described in U.S. Patent Application Publication No. 2010/0300571, which is incorporated herein by reference in its entirety. For example, to reduce the permeability of the polymeric material-based oxygen barrier layer, it is possible to add lamellar nanofillers to the polymeric material matrix. Such a reduction in permeability may be attributed to an effect of "tortuosity" brought about by the lamellar nanofillers. This is because the oxygen has to follow a much longer pathway because of these obstacles arranged in successive strata. Theoretical models regard the barrier effects as becoming more pronounced as the aspect ratio, that is to say the length/thickness ratio, increases.

The lamellar nanofillers which are most widely investigated today are clays of smectite type, mainly montmorillonite. The difficulty of use lies first of all in the more or less extensive separation of these individual lamellae, that is to say the exfoliation, and in their distribution, in the polymer. To help in the exfoliation, use may be made of an "intercalation" technique, which consists in swelling the crystals with organic cations, generally quaternary ammonium cations, which will compensate for the negative charge of the lamellae. These crystalline aluminosilicates, when they are exfoliated in a thermoplastic matrix, exist in the form of individual lamellae, the aspect ratio of which may reach values of the order of 500 or more.

In accordance with another aspect of the present invention, the inorganic additive may include particles based on zirconium, titanium, cerium and/or silicon phosphate, in the form of non-exfoliated nanometric lamellar compounds, as disclosed for example in U.S. Patent Application Publication No. 2007/0082159, the relevant portions of which are hereby incorporated herein by reference.

The inorganic additive content of the polymeric material used in constructing the oxygen barrier layer **203** may vary depending on the desired level of reduction in oxygen permeability. When present, the inorganic additive may be present in the polymeric material in an amount from 0.01% to about 50% by weight with respect to the total weight of the oxygen barrier layer **203** composition.

Turning to FIGS. 6A and 6B, a barrier inner tube **608**, which includes a combination of a polymeric material and one or more of the inorganic additives described above, may

be used in constructing the hot melt adhesive transfer hose **600**. The inorganic additive content of the polymeric material used in constructing the barrier inner tube **608** may vary depending on the desired level of reduction in oxygen permeability. The inorganic additive may be present in the polymeric material in an amount from 0.01% to about 50% by weight with respect to the total weight of the barrier inner tube **608** composition. In one non-limiting example, the barrier inner tube **608** may comprise a fluoropolymer (e.g., polytetrafluoroethylene) in combination with a sufficient quantity of one or more of the inorganic additives described above to provide the desired level of reduction in oxygen permeability.

With respect to placement of the oxygen barrier layer **203** outside the hot zone (e.g., outside the insulation sublayer of outer layer **106** or outside the protective sublayer of outer layer **106**), any airtight layer will expand and contract as the gases trapped within the insulation layers expand under the heat applied by the heating tape. To accommodate any thermal expansion, the oxygen barrier layer **203** may be oversized and/or corrugated, or fitted with a one-way valve **204** (shown in FIGS. **5A** and **5B**) to allow the expansion gases to escape and subsequently blocks the ingress of oxygen. As noted above, placement of the oxygen barrier layer **203** outside the hot zone further permits lower melting or softening materials to be used for this purpose. For example, metalized polymer films (e.g., aluminized mylar) may be used outside the hot zone, in addition to the higher melting materials discussed above.

Although not shown, it should be appreciated that the transfer hose **600** shown in FIGS. **6A** and **6B** may also include a distinct oxygen barrier layer **203**, such as that shown in FIGS. **2-5**. It should be further appreciated that while the oxygen barrier layer **203** or the barrier inner tube **608** may block or inhibit the ingress of oxygen into the

conduit transporting the hot melt adhesive, it may be advantageous to utilize a low oxygen or inert gas atmosphere during the manufacture of the oxygen barrier layer **203**, the barrier inner tube **608**, and/or the hot melt adhesive transfer hose.

EXAMPLES

Standard: To qualitatively evaluate the observed discoloration of Henkel 614C hot melt adhesive in hot melt adhesive transfer tubing, twelve (12) samples of Henkel 614C contained in aluminum sample pans heated in a laboratory oven at 350° F. under ambient atmosphere from 0 to about 72 hours. At various intervals, a sample was removed from the oven and allowed to cool to room temperature. The degree of discoloration increases with increased residence time in the 350° F. oven (see FIG. **7**). Based on the illustrations of the 12 samples, FIG. **8** illustrates a qualitative color scale ranging from a base line of 1 (for an un-heated sample of Henkel 614C) to a maximum of 12 (for a sample heated for about 95 hours).

Comparative testing: Multiple samples were evaluated using different tubing with or without any oxygen barrier layers applied. As shown in FIG. **9**, different tubing types include 0.030" PTFE (Dupont Teflon® 62X resin based) core with or without stainless steel braid; 0.040" PTFE core with or without stainless steel braid; or 0.030" PTFE carbon-lined core with or without stainless steel braid. As shown in FIG. **10**, the tubing samples without stainless steel braid were sealed using black plastic screw caps. Multiple tubing and aluminum pan ("ratpan") samples were prepared as described in Table 2 (below), and placed in a laboratory oven (see FIG. **11**) and heated to 350° F. for 29 hours. The samples were qualitatively evaluated using the developed color scale described above.

TABLE 2

Henkel 614C test samples and results.					
#	Description	L	Misc	Test	Color Scale
1a	0.040" PTFE core	8-3/8"	no wrap	Put in oven at 350° F. 2/23 10:30 am; removed at 29 hrs; let cool overnight	10/11 (light side/dark side)
1b	0.040" PTFE core w/silicone tape	6-3/8"	2 layers of 272864, 1/2 overlap, wrapped in same direction	Put in oven at 350° F. 2/23 10:30 am; removed at 29 hrs; let cool overnight	9/10
1c	0.040" PTFE core w/foil	7-6/8"	approx 4 layers of household aluminum foil, also some aluminum tape in an attempt to seal it	Put in oven at 350° F. 2/23 10:30 am; removed at 29 hrs; let cool overnight	6-7/8-10 in middle, 3-5/6 on ends
2	0.040" PTFE core w/carbon	2 ft		Put in oven at 350° F. 2/23 10:30 am; removed at 29 hrs; let cool overnight	9/10
3	0.030" PTFE core	2 ft		Put in oven at 350° F. 2/23 10:30 am; removed at 29 hrs; let cool overnight	10/11
4	Hose, 0.040" PTFE core w/stainless steel ("ss") braid	2 ft	poured out 12.5 g	Put in oven at 350° F. 2/23 10:30 am; poured out at 29 hrs	9-10

TABLE 2-continued

Henkel 614C test samples and results.					
#	Description	L	Misc	Test	Color Scale
5	Hose, 0.040" PTFE carbon-lined core w/ss braid	2 ft	poured out 9.3 g	Put in oven at 350° F. 2/23 10:30 am; poured out at 29 hrs	9
6	Hose, 0.030" PTFE core w/ss braid	2 ft	poured out 9.3 g	Put in oven at 350° F. 2/23 10:30 am; poured out at 29 hrs	10-11
1	Ratpan of 614C		≈19 g		
2	Ratpan of 614C		≈4.0 g	Put in oven at 350° F. 2/23 10:30 am; removed at 24 hrs	12
3	Ratpan of 614C		≈6.9 g	Put in oven at 350° F. 2/23 10:30 am; removed at 24 hrs	12
4	Ratpan of 614C		≈13.0 g	Put in oven at 350° F. 2/23 10:30 am; removed at 24 hrs	11
5	Ratpan of 614C		≈21.1 g	Put in oven at 350° F. 2/23 10:30 am; removed at 29 hrs	9
6	Ratpan of 614C		≈25.3 g	Put in oven at 350° F. 2/23 10:30 am; removed at 29 hrs	9
7	Ratpan of 614C		≈30.0 g	Put in oven at 350° F. 2/23 10:30 am; removed at 29 hrs	9

As illustrated in FIG. 12, the aluminum foil-wrapped PTFE inner tube (Sample 1c) provided an improvement in the discoloring of the molten hot melt adhesive under simulated testing conditions as compared to the uncoated PTFE tube (Sample 1a) or the silicone tape wrapped PTFE tube (Sample 1b). Although not shown, significant discoloring (e.g., 9 or higher) was still observed with the PTFE tubing that had an interior carbon coating. Further, it should be understood that in the event that color evaluation of the molten hot melt adhesive is to be monitored in-process, the equipment and methods described in commonly-assigned U.S. Patent Application Publication No. 2014/0144933, which is expressly incorporated herein by reference in its entirety, may be utilized.

Metal Barrier Layer

In accordance with another aspect of the present disclosure, there is provided a multi-layered hot melt adhesive transfer hose 700 configured to prevent air and other gasses from penetrating the hose and contacting the hot melt adhesive therein. As shown in FIGS. 13A and 13B, the hot melt adhesive transfer hose 700 includes a barrier layer comprising an inner tube 702 configured to prevent the ingress of oxygen and other gases into the conduit of the hose, thereby eliminating discoloration and any associated degradation of the molten hot melt adhesive. Further, the prevention of oxygen passing into the hose also preserves the expected "pot life" of the hot melt adhesive.

The inner tube 702 forms the operative core of the hose 700 through which the molten hot melt adhesive actually flows. The inner tube 702 is preferably made from a metallic material capable of withstanding relatively high temperatures such as stainless steel. In another aspect of the disclosure, the barrier layer can be a metal tape, such as aluminum, or a metallic coating applied to the core as will be discussed in greater detail below. Unlike an inner tube of conventional hoses, the metal inner tube 702 of the hot melt adhesive hose 700 is gas impermeable, and therefore does not allow any air

to diffuse into the conduit containing the hot melt adhesive. Thus, the barrier layer comprising the metal inner tube 702 prevents oxygen and other gasses from entering the hose and coming into contact with the hot melt adhesive, thereby eliminating charring.

Additionally, the metal inner tube 702 blocks moisture from affecting hot melt PURs, and can also keep gasses within the hose from escaping. For instance, foamed hot melt within the metal inner tube 702 maintains gas in solution longer than standard hoses. Moreover, manufacturing an imperviable flexible barrier layer comprising a metal inner tube according to the present disclosure is more time and cost efficient since existing hose manufacturing processes can be utilized and/or easily modified.

The inner tube 702 has seamless corrugations to allow for flexibility, and preferably maintains a consistent tube wall thickness throughout the hose to minimize residual stress in the metal. Further, the inner tube 702 may be oversized. The inner tube 702 is reinforced by a reinforcing layer, or a structural layer 704 to help withstand the high fluid pressure used to transfer the molten hot melt adhesive. The structural layer 704, which is disposed about the exterior of the inner tube 702 and serves to provide added strength and protection to the inner tube, may comprise a braided jacket of thermally stable material.

Accordingly, the structural layer 704 serves to provide desired physical integrity of the hot melt adhesive transfer hose. In one non-limiting example, the structural layer 704 includes a braided jacket made of stainless steel. The braiding frequency and/or thickness of the braid may vary depending on the expected temperature and pressure limits of the hot melt adhesive apparatus. The braid may be a spiral, a helical, woven/interwoven, or hoop/loop patterns such as square or single braids. Further, the hose 700 can include a plurality of structural layers 704 to provide additional reinforcement for high pressure capability. As shown in FIG. 14, a hose 700 having two overlapping structural layers 704, 705 is provided.

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Additionally, the hot melt adhesive transfer hose **700** may comprise a liner **710** provided on an inner surface of the inner tube **702** as shown in FIGS. **15A** and **15B**. The liner **710** preferably comprises a heat stable polymeric material such as polytetrafluoroethylene (PTFE) or other polymer capable of withstanding relatively high temperatures. In one aspect of the transfer hose **700**, the liner **710** is a distinct layer that is in contact with an inner surface of the corrugated inner tube **702**. The liner is configured to improve flow of the hot melt adhesive by minimizing material incompatibility with the inner tube **702**. In particular, the liner **710** has a smooth interior surface to facilitate fluid flow and prevent material incompatibility issues between the transferred media and the metal inner tube **702**. For instance, the liner **710** prevents material from becoming lodged within grooves of the interior corrugations of the inner tube **702**.

The hose **700** may further comprise an outer covering **706**. The outer covering **706** may be a multi-layered structure having a heating tape sublayer, an insulation sublayer, and/or a protective sublayer, which are not shown. The heating tape serves to provide uniform conduction heating to the hot melt adhesive hose. The insulation layer, which may include but is not limited to aramid felt insulation or fiber glass insulation, serves to control heat loss and may also improve chemical and moisture resistance. The protective layer, which principally serves to protect the hose and other inner component/layers from damage caused by physical abuse, may include, but is not limited to, a durable polymer such as aramid sheathing.

Each sublayer of the outer covering is a separate and distinct layer. For instance, the heating tape sublayer is in contact with an outer surface of the structural layer **704** and is covered by the insulation sublayer. The protective sublayer covers the insulation sublayer.

The inner tube **702**, the structural layer **704**, and/or the heat tape sublayer of the outer covering **706** are in what may be referred to as a "hot zone," which is near or above the temperature of the molten hot melt adhesive. Alternatively, the outer surface of the insulation or the protective sublayers of the outer layer **706** are radially disposed from and are outside the hot zone, and are thus subjected to a lower temperature.

Further, as schematically represented in FIG. **17**, the transfer hose **700** can be used in a hot melt system **720** that includes a dispenser **722** for dispensing hot melt adhesive onto a substrate. The hose **700** may be operatively coupled to the dispenser **722**. The hose may also be operatively coupled to a source **724** of liquid hot melt adhesive, such as a melter.

As previously noted above, hot melt adhesives are heated to temperatures sufficient to melt the solid form to a molten, flowable state, which are generally in a range from about 100° C. (about 212° F.) to about 230° C. (about 450° F.). In particular, the transfer hose can further include a heater **750**, schematically represented in FIG. **18**, for maintaining the hot melt adhesive at a set point. The heater can include at least one heating wire comprising metal such as copper, aluminum, or silver. The at least one heating wire may circumferentially wrap around an exterior surface of the inner core of the hose in a helical pattern, as schematically represented in FIG. **18**. Alternatively, the at least one heating wire may be provided on an exterior surface of the inner core in a direction parallel to the longitudinal axis of the inner core. The heater **750** may be electrically coupled to a power source **752**, as schematically represented in FIG. **18**.

The heater is adapted to be electrically coupled to a power source controlled to maintain the set point to about 450° F.

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or less for an extended period of time. For example, the flexible metal tube may further transport the hot melt adhesive at a set point temperature at or below about 400° F., at a set point temperature at or below about 350° F., or at a set point temperature at or above about 250° F. A sensor can further be provided on the heater, or alternatively, within the hose, for measuring the set point temperature.

Additionally, it was observed that the metal barrier layer **702** unexpectedly prevented the hot melt adhesive from charring within the hose **700** at certain temperatures for extended periods of time. Specifically, an unknown effect of the barrier layer **702** was observed, in which no charring of hot melt adhesive occurred for at least ninety-six hours when the adhesive was heated to a set point temperature in the range of about or greater than 250° F. up to and including about 450° F.

Referring to FIG. **16**, an example of the observed discoloration over time of common hot melt adhesive (such as Technomelt® Supra 614C by Henkel) in a competitor's hot melt adhesive transfer hose is shown in line A, as compared to the observed discoloration over time of the hot melt adhesive in the transfer hose **700** of the present disclosure is shown in line B. As a comparative standard, the hot melt adhesive was maintained at a setpoint of 350° F. in both the competitor's transfer hose and the present disclosure's transfer hose **700**. Further, the color of the adhesive was observed every eight hours over a period of ninety-six total hours.

As illustrated, the extent of discoloration of hot melt adhesive in the competitor's transfer hose increased greatly over time. For example, the sample of adhesive **A9** in the competitor's transfer hose at ninety-six hours was significantly discolored and contained a substantial amount of char. Conversely, the sample of adhesive **B9** in the transfer hose of the present disclosure at ninety-six hours was not significantly discolored and contained no significant amount of char formation. As illustrated, the sample of adhesive **B9** in the transfer hose of the present disclosure at ninety-six hours had substantially the same color as the sample of adhesive **B1** in the transfer hose of the present disclosure at startup.

Moreover, as shown in FIG. **16**, an unknown effect of the hose **700** is that there is no significant discoloration and charring of adhesive at 350° F. within the hose at eight hours (**B2**), sixteen hours (**B3**), twenty-four hours (**B4**), thirty-two hours (**B5**), forty hours (**B6**), forty-eight hours (**B7**), seventy-two hours (**B8**) and ninety-six hours (**B9**). More particularly, the metal barrier layer **702** unexpectedly resulted in less than 1.0% char formation at 24 hours. Similarly, use of the metal barrier layer **702** unexpectedly resulted in less than 1.0% char formation at 48 hours, at 72 hours, and even at 96 hours, as shown in FIG. **16**. Conversely, as shown in FIG. **16**, the adhesive inside the competitor's hose displayed significant discoloration and charring at eight hours (**A2**). The severity of discoloration and charring within the competitor's hose worsened at sixteen hours (**A3**), twenty-four hours (**A4**), thirty-two hours (**A5**), forty hours (**A6**), forty-eight hours (**A7**), seventy-two hours (**A8**), and ninety-six hours (**A9**).

FIG. **19** schematically illustrates an apparatus for melting and dispensing a hot melt adhesive **760**. The apparatus **760** comprises a chamber **762** for receiving a solid form of the hot melt adhesive, a heating device **764** coupled to the chamber **762** and configured to receive the hot melt adhesive for liquefying the solid form of the hot melt adhesive, and the multi-layered hot melt adhesive transfer hose **700** fluidly coupled to the heating device **764**.

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While the present disclosure has been illustrated by the description of specific embodiments thereof, and while the embodiments have been described in considerable detail, it is not intended to restrict or in any way limit the scope of the appended claims to such detail. The various features discussed herein may be used alone or in any combination. The present disclosure is therefore not limited to the specific details, representative apparatus and methods and illustrative examples shown and described. Instead, the present disclosure is intended to cover such alternatives, modifications, and equivalents without departing from the spirit and scope of the application as defined by the appended claims.

What is claimed is:

1. A multi-layered hot melt adhesive transfer hose, comprising:

a flexible metal inner tube for transporting heated liquid hot melt adhesive and forming an impermeable barrier layer configured to prevent an ingress of oxygen from passing through the flexible metal inner tube and into the hot melt adhesive;

at least one structural layer configured to withstand a fluid pressure;

an outer covering overlaying and in contact with an exterior surface of the at least one structural layer; and a heater for maintaining the hot melt adhesive at a set point, wherein the heater is adapted to be electrically coupled to a power source controlled to maintain the set point to about 230° C. (450° F.) or less for an extended period of time.

2. The hose of claim 1, wherein the flexible metal inner tube comprises a plurality of seamless corrugations.

3. The hose of claim 1, wherein the flexible metal inner tube is a metal tape and/or comprises aluminum.

4. The hose of claim 2, further comprising a liner disposed along an inner surface of the flexible metal inner tube, wherein the liner comprises a heat stable polymeric material configured to improve flow of the heated liquid hot melt adhesive through the flexible metal inner tube, and/or wherein the liner includes a smooth interior surface to facilitate fluid flow and prevent material incompatibility between the heated liquid hot melt adhesive and the flexible metal inner tube.

5. The hose of claim 1, wherein the at least one structural layer comprises two overlapping braided jackets of thermally stable material, and wherein the two overlapping braided jackets that are configured to withstand the fluid pressure and are disposed about an exterior of the flexible metal inner tube.

6. The hose of claim 1, wherein the outer covering has a multilayered structure, wherein the outer covering further comprises a heating tape sublayer, an insulation sublayer covering the heating tape sublayer, and a protective sublayer covering the insulation sublayer.

7. The hose of claim 1, wherein the hose is configured to transfer the hot melt adhesive, wherein the flexible metal inner tube forms a conduit for transporting the hot melt adhesive, wherein the flexible metal inner tube forms an operative core of the hose through which the hot melt adhesive flows while in a molten state, and wherein the flexible metal inner tube is made from stainless steel and comprises a plurality of seamless corrugations.

8. The hose of claim 7, wherein either

1) The outer covering comprises a multilayered structure that comprises a heating tape sublayer, an insulation sublayer covering the heating tape sublayer, and a protective sublayer covering the insulation sublayer; or

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2) The outer covering comprises a multilayered structure that comprises a heating wire sublayer, an insulation sublayer covering the heating wire sublayer, and a protective sublayer covering the insulation sublayer, wherein the heating wire sublayer comprises at least one heating wire comprising metal, and wherein the at least one heating wire circumferentially wraps around an exterior surface of the operative core of the hose in a helical pattern, and wherein the heater comprises the heating wire.

9. The hose of claim 7, wherein the plurality of seamless corrugations is a plurality of helical seamless corrugations, and wherein the hose is integrally formed.

10. The hose of claim 9, wherein the at least one structural layer comprises two overlapping braided jackets of thermally stable material, and wherein the two overlapping braided jackets that are configured to withstand the fluid pressure and are disposed about an exterior of the flexible metal inner tube and configured to provide added strength and protection to the inner tube such that the hose is configured to withstand fluid pressures of up to 10.3 MPa (1500 psi).

11. The hose of claim 10, wherein either

1) The outer covering comprises a multilayered structure that comprises a heating tape sublayer, an insulation sublayer covering the heating tape sublayer, and a protective sublayer covering the insulation sublayer; or

2) The outer covering comprises a multilayered structure that comprises a heating wire sublayer, an insulation sublayer covering the heating wire sublayer, and a protective sublayer covering the insulation sublayer, wherein the heating wire sublayer comprises at least one heating wire comprising metal, and wherein the at least one heating wire circumferentially wraps around an exterior surface of the operative core of the hose in a helical pattern, and wherein the heater comprises the heating wire.

12. The hose of claim 1, wherein the at least one structural layer is configured to withstand fluid pressures of up to 10.3 MPa (1500 psi), and wherein the at least one structural layer comprises two overlapping braided jackets of thermally stable material, and wherein the two overlapping braided jackets that are configured to withstand the fluid pressure and are disposed about an exterior of the flexible metal inner tube such that the hose is configured to withstand fluid pressures of up to 10.3 MPa (1500 psi).

13. An apparatus for melting and dispensing a hot melt adhesive, the apparatus comprising:

a chamber for receiving a solid form of the hot melt adhesive;

a heating device coupled to said chamber and configured to receive the hot melt adhesive for liquefying the solid form of the hot melt adhesive; and

the multi-layered hot melt adhesive transfer hose of claim 1 fluidly coupled to the heating device.

14. The apparatus of claim 13, further comprising the power source,

wherein the set point is in a range between 121.1° C. (250° F.) to 177° C. (350° F.),

wherein the hose is configured to transfer the hot melt adhesive, wherein the flexible metal inner tube forms a conduit for transporting the hot melt adhesive, wherein the flexible metal inner tube forms an operative core of the hose through which the hot melt adhesive flows while in a molten state, and wherein the flexible metal inner tube is made from stainless steel and comprises a plurality of helical seamless corrugations,

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wherein the at least one structural layer comprises two overlapping braided jackets of thermally stable material, and wherein the two overlapping braided jackets that are configured to withstand the fluid pressure and are disposed about an exterior of the flexible metal inner tube and configured to provide added strength and protection to the inner tube such that the hose is configured to withstand fluid pressures of up to 10.3 MPa (1500 psi), and

wherein either

- 1) The outer covering comprises a multilayered structure that comprises a heating tape sublayer, an insulation sublayer covering the heating tape sublayer, and a protective sublayer covering the insulation sublayer; or
- 2) The outer covering comprises a multilayered structure that comprises a heating wire sublayer, an insulation sublayer covering the heating wire sublayer, and a protective sublayer covering the insulation sublayer, wherein the heating wire sublayer comprises at least one heating wire comprising metal, and wherein the at least one heating wire circumferentially wraps around an exterior surface of the operative core of the hose in a helical pattern, and wherein the heater comprises the heating wire.

15. A hot melt system, comprising:

a dispenser for dispensing hot melt adhesive onto a substrate; and

the multi-layered hot melt adhesive transfer hose of claim 1 operatively coupled to a source of liquid hot melt adhesive and the dispenser.

16. A method of dispensing hot melt adhesive, comprising:

melting hot melt adhesive;

transporting the melted hot melt adhesive through a hose to a dispenser, the hose being a multi-layered hot melt adhesive transfer hose of claim 1;

heating the melted hot melt adhesive in the hose at the set point of about 230° C. (450° F.) or less;

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preventing, with a barrier layer, transfer of oxygen into the adhesive within the hose; and
dispensing the hot melt adhesive onto a substrate.

17. The method of claim 16, wherein the set point is about 177° C. (350° F.) or less.

18. The method of claim 16, wherein the set point is in a range between 121.1° C. (250° F.) to 177° C. (350° F.), wherein the hose transfers the hot melt adhesive, wherein the flexible metal inner tube forms a conduit for transporting the hot melt adhesive, wherein the flexible metal inner tube forms an operative core of the hose through which the hot melt adhesive flows while in a molten state, and wherein the flexible metal inner tube is made from stainless steel and comprises a plurality of helical seamless corrugations.

19. The method of claim 18, wherein the at least one structural layer comprises two overlapping braided jackets of thermally stable material, and wherein the two overlapping braided jackets that are configured to withstand the fluid pressure and are disposed about an exterior of the flexible metal inner tube and configured to provide added strength and protection to the inner tube such that the hose is configured to withstand fluid pressures of up to 10.3 MPa (1500 psi).

20. The method of claim 18, wherein either

- 1) The outer covering comprises a multilayered structure that comprises a heating tape sublayer, an insulation sublayer covering the heating tape sublayer, and a protective sublayer covering the insulation sublayer; or
- 2) The outer covering comprises a multilayered structure that comprises a heating wire sublayer, an insulation sublayer covering the heating wire sublayer, and a protective sublayer covering the insulation sublayer, wherein the heating wire sublayer comprises at least one heating wire comprising metal, and wherein the at least one heating wire circumferentially wraps around an exterior surface of the operative core of the hose in a helical pattern, and wherein the heater comprises the heating wire.

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